

Any of these factors or all of these factors could cause the operating point to drift from the desired location as fixed by the selected values V_{CE} and I_C in the output characteristics. Typical variations of these parameters at different temperatures for a silicon transistor are shown in Table 2.2.

Table 2.2 Typical variations of I_{CO} , V_{BE} and β with temperature

Temperature	I_{CO}	V_{BE}	β
– 65° C	0.0002 nA	0.85 V	20
25° C	0.1 nA	0.65 V	50
100° C	20 nA	0.48 V	80
175° C	3300 nA	0.3 V	120

We know that the collector current for the common emitter configuration is given by

$$I_C = \beta I_B + (\beta + 1) I_{CO} \quad (2.3)$$

As I_{CO} increases with temperature, I_C also increases resulting in a further increase in temperature which in turn increases β . This further increases I_C and the resulting cumulative effect causes the operating point to drift into the saturation region.

◆ 2.4 STABILITY FACTORS

Transistor circuits must be designed to provide a certain extent of temperature stability so that changes in temperature result in minimal changes in operating point. The maintenance of the operating point at the desired value is specified by stability factors, which represents the extent of change of operating point due to variation in temperature.

A stability factor, can be defined for each of the parameters that affect bias stability as follows:

$$S(I_{CO}) = \frac{\partial I_C}{\partial I_{CO}} \quad \text{at constant } \beta \text{ and } V_{BE}$$

$$\text{or} \quad S(I_{CO}) = \left. \frac{\partial I_C}{\partial I_{CO}} \right|_{\beta, V_{BE} \text{ constant}} \quad (2.4)$$

$$\text{Similarly,} \quad S(V_{BE}) = \left. \frac{\partial I_C}{\partial V_{BE}} \right|_{I_{CO}, \beta \text{ constant}} \quad (2.5)$$

$$\text{and} \quad S(\beta) = \left. \frac{\partial I_C}{\partial \beta} \right|_{I_{CO}, V_{BE} \text{ constant}} \quad (2.6)$$

Observe from these equations that, for circuits with low stability factors, change in collector current ΔI_C is small when I_{CO} , V_{BE} and β changes with temperature. Hence circuits with low stability factors are relatively stable and insensitive to variations in temperature.

Total Change in Collector Current

From Equations (2.4) to (2.6) we can write an expression for the total change in the collector current as

$$\Delta I_C = \frac{\partial I_C}{\partial I_{CO}} \Delta I_{CO} + \frac{\partial I_C}{\partial V_{BE}} \Delta V_{BE} + \frac{\partial I_C}{\partial \beta} \Delta \beta \quad (2.7)$$

$$\text{or} \quad \Delta I_C = S(I_{CO}) \Delta I_{CO} + S(V_{BE}) \Delta V_{BE} + S(\beta) \Delta \beta \quad (2.8)$$

Lower stability factors therefore imply lower variations in the collector current.

◆ 2.5 FIXED-BIAS CIRCUIT

There are several circuit configurations through which the transistor current and voltage can be adjusted in order to fix the operating point. Fixed-bias, emitter-bias or self-bias and voltage divider bias are some common schemes. In this section, the fixed bias circuit is presented. Figure 2.4 shows a typical fixed-bias circuit.

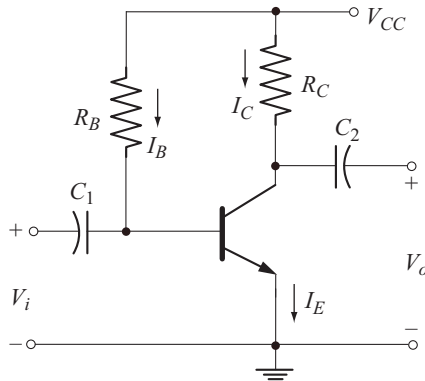


Fig. 2.4 Fixed-bias circuit

The dc equivalent circuit can be written by replacing the capacitors with open circuits (since capacitors block dc) as shown in Fig. 2.5.

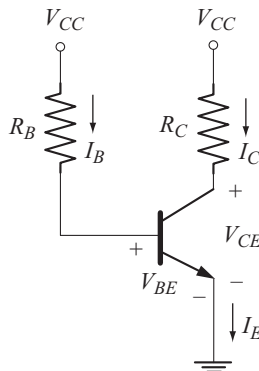


Fig. 2.5 DC equivalent circuit of the fixed-bias circuit

Writing Kirchhoff's voltage law equation to the base-emitter circuit, we get,

$$V_{CC} = I_B R_B + V_{BE}$$

or

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \quad (2.9)$$

Now let us look at the collector-emitter circuit. The magnitudes of the collector current and base current neglecting I_{CO} are related by

$$I_C = \beta I_B \quad (2.10)$$

where β is the dc current gain in common-emitter configuration. Observe from Equation (2.9), that I_B is decided by R_B while I_C is decided by β . Thus, R_C has no role in the value of I_C when the transistor is the active region. However, the value of I_C determines the value of V_{CE} .

Writing Kirchhoff's voltage law equation to the collector-emitter circuit of Fig. 2.5, we get

$$V_{CC} = I_C R_C + V_{CE} \quad (2.11)$$

or

$$V_{CE} = V_{CC} - I_C R_C \quad (2.12)$$

Transistor in Saturation

Figure 2.6 shows the transistor operating in the saturation region.

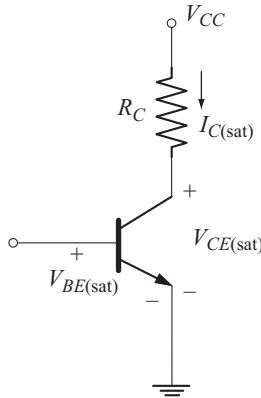


Fig. 2.6 Transistor in saturation

From Equation (2.11), the collector current in saturation, $I_{C(sat)}$ is given by

$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C} \quad (2.13)$$

But

$$V_{CE(sat)} \approx 0$$

$\therefore$

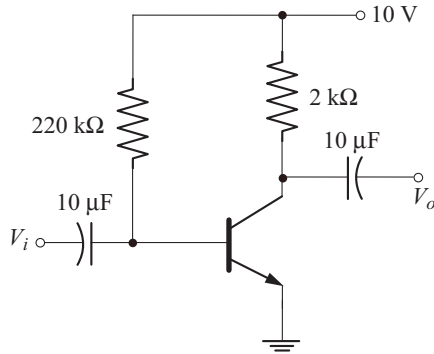
$$I_{C(sat)} \approx \frac{V_{CC}}{R_C} \quad (2.14)$$

Note that in the saturation region I_C is solely decided by R_C .

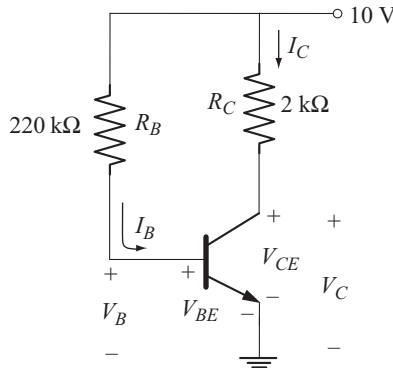
Example 2.1

For the fixed-bias circuit shown, assuming $V_{BE} = 0.7\text{ V}$ and $\beta = 60$ find

- Quiescent values of base and collector currents
- Quiescent value of V_{CE}
- Base-ground and collector-ground voltages
- Base-collector voltage
- Quiescent values of I_C and V_{CE} for $\beta = 110$

**Solution**

Treating the coupling capacitors as open circuits, let us mark the various currents and voltages in the circuit.



- Quiescent values of base and collector currents

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{10\text{ V} - 0.7\text{ V}}{220\text{ k}\Omega} = 42.27\text{ }\mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = 60 \times 42.27\text{ }\mu\text{A} = 2.54\text{ mA}$$

- Quiescent value of V_{CE}

$$V_{CC} = I_C R_C + V_{CE}$$

$\therefore$

$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10 - (2.54\text{ mA})(2\text{ k}\Omega) = 4.92\text{ V}$$

(c) Base and collector voltages with respect to ground

$$V_B = V_{BE} = 0.7 \text{ V}$$

$$V_C = V_{CE} = 4.92 \text{ V}$$

(d) Base-collector voltage

$$V_{BC} = V_B - V_C = 0.7 \text{ V} - 4.92 \text{ V} = -4.22 \text{ V}$$

(e) When $\beta = 110$

I_{BQ} is not affected by change in β .

$$\therefore I_{BQ} = 42.27 \mu\text{A} \quad [\text{as obtained in part (a)}]$$

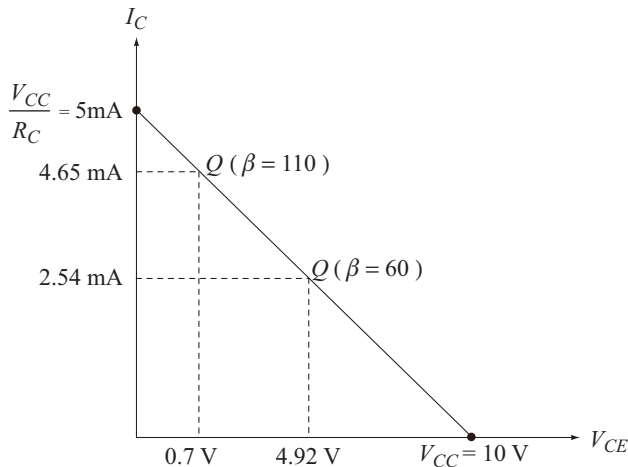
$$\therefore I_{CQ} = \beta I_{BQ} = 110 \times 42.27 \mu\text{A} = 4.65 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10 \text{ V} - (4.65 \text{ mA} \times 2 \text{ k}\Omega) = 0.7 \text{ V}$$

Let us now tabulate the Quiescent point for $\beta = 60$ and $\beta = 110$

β	I_{BQ}	I_{CQ}	V_{CEQ}
60	42.27 μA	2.54 mA	4.92 V
110	42.27 μA	4.65 mA	0.7 V

The Q points are indicated on the dc load line in the following figure.



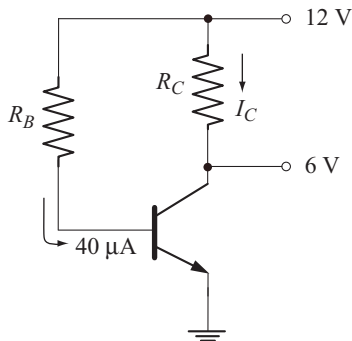
Observe that, when β changes from 60 to 110, the Q point shifts from the middle of the active region to near saturation. Therefore fixed bias circuit has very poor stability of the operating point. The reason for this is that, there is no change in the value of the base current.

Example 2.2

For the fixed-bias circuit shown, determine

- (a) Collector current, I_C
- (b) Collector resistance, R_C
- (c) Base resistance, R_B
- (d) V_{CE}

Assume $\beta = 80$ and $V_{BE} = 0.7$ V.

**Solution**

- (a) Collector current

$$I_C = \beta I_B = (80)(40 \mu\text{A}) = 3.2 \text{ mA}$$

- (b) Collector resistance

Given

$$V_C = 6 \text{ V}$$

$$V_{CE} = V_C = 6 \text{ V}$$

$$R_C = \frac{V_{CC} - V_{CE}}{I_C} = \frac{12 \text{ V} - 6 \text{ V}}{3.2 \text{ mA}} = 1.875 \text{ k}\Omega$$

- (c) Base resistance

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{40 \mu\text{A}} = 282.5 \text{ k}\Omega$$

- (d) V_{CE}

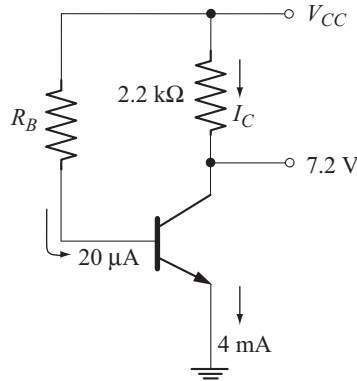
$$\begin{aligned} V_{CE} &= \text{Voltage from collector to emitter} \\ &= V_C = 6 \text{ V.} \end{aligned}$$

Example 2.3

For the circuit shown, find

- (a) I_C
- (b) V_{CC}
- (c) β
- (d) R_B

Assume $V_{BE} = 0.7$ V.

**Solution**

$$I_B = 20 \mu\text{A} \quad I_E = 4 \text{ mA} \quad V_{CE} = 7.2 \text{ V}$$

$$V_{BE} = 0.7 \text{ V} \quad R_C = 2.2 \text{ k}\Omega$$

We are required to find I_C and V_{CC} .

$$(a) \quad I_C + I_B = I_E$$

$$\therefore \quad I_C = I_E - I_B = 4 \text{ mA} - 0.02 \text{ mA} = 3.98 \text{ mA}$$

$$(b) \quad V_{CC} = I_C R_C + V_{CE}$$

$$= (3.98 \text{ mA} \times 2.2 \text{ k}\Omega) + 7.2 \text{ V}$$

$$= 15.956 \text{ V}$$

$$\text{Let} \quad V_{CC} = 16 \text{ V}$$

$$(c) \quad \beta = \frac{I_C}{I_B} = \frac{3.98 \text{ mA}}{0.02 \text{ mA}} = 199$$

$$(d) \quad R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{16 \text{ V} - 0.7 \text{ V}}{0.02 \text{ mA}} = 765 \text{ k}\Omega$$

Example 2.4

- (a) For the fixed-bias circuit, $R_B = 50 \text{ k}\Omega$, $R_C = 500 \Omega$, $V_{CC} = 10 \text{ V}$. Find the coordinates of the operating point. Draw the dc load line and locate the operating point on the dc load line. Assume silicon transistor with $\beta = 50$ and $V_{BE} = 0.7 \text{ V}$.
- (b) Find the new operating point if the transistor is replaced by a similar transistor with $\beta = 100$. Comment on the stability of the circuit.

Solution

$$(a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{10 \text{ V} - 0.7 \text{ V}}{50 \text{ k}\Omega} = 0.186 \text{ mA}$$

$$I_C = \beta I_B = 50 \times 0.186 \text{ mA} = 9.3 \text{ mA}$$

Writing KVL equation for the collector circuit,

$$\begin{aligned} V_{CC} &= I_C R_C + V_{CE} \\ \therefore V_{CE} &= V_{CC} - I_C R_C \\ &= 10 \text{ V} - (9.3 \text{ mA} \times 0.5 \text{ k}\Omega) = 5.35 \text{ V} \end{aligned}$$

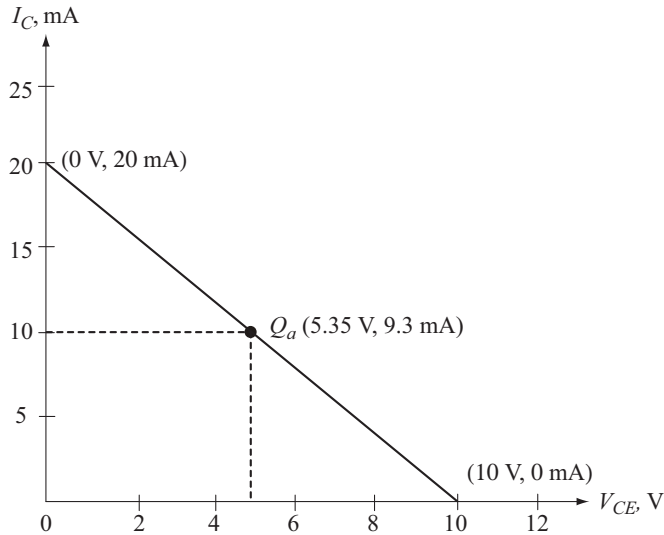
Therefore the operating point is at $(V_{CE}, I_C) = (5.35 \text{ V}, 9.3 \text{ mA})$

$$\text{Let } Q_a = (5.35 \text{ V}, 9.3 \text{ mA})$$

The current axis intercept of the load line is

$$\frac{V_{CC}}{R_C} = \frac{10 \text{ V}}{0.5 \text{ k}\Omega} = 20 \text{ mA}$$

and the voltage axis intercept is $V_{CC} = 10 \text{ V}$. The load line is plotted by joining $(0 \text{ V}, 20 \text{ mA})$ and $(10 \text{ V}, 0 \text{ mA})$ as shown below:



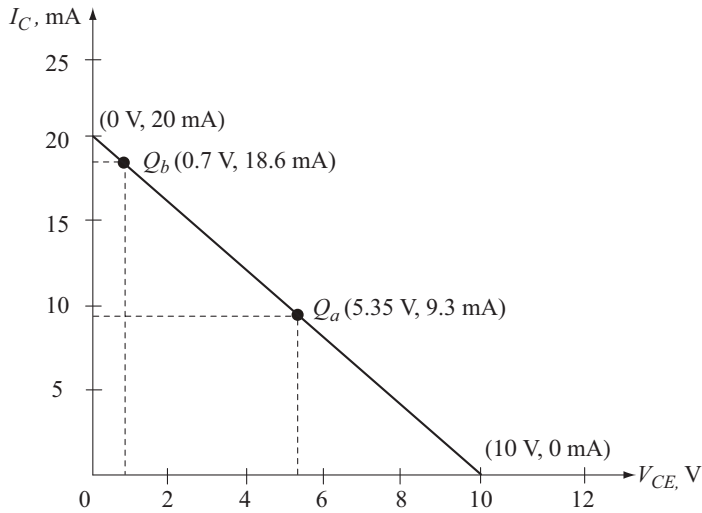
The Q -point is nearly at the center of the load line.

- (b) Now, let us locate the new position of the operating point when a transistor with $\beta = 100$ is used.

The base current is obviously not affected and remains at $I_B = 0.186 \text{ mA}$.

$$\begin{aligned} I_C &= \beta I_B = 100 \times 0.186 \text{ mA} = 18.6 \text{ mA} \\ \text{and } V_{CE} &= V_{CC} - I_C R_C = 10 \text{ V} - (18.6 \text{ mA})(0.5 \text{ k}\Omega) = 0.7 \text{ V} \end{aligned}$$

The Q -point now shifts to $Q_b = (0.7 \text{ V}, 18.6 \text{ mA})$.



The Q -point has moved into the saturation region. In general, fixed-bias circuits do not provide stability against variation in β .

◆ 2.6 LOAD LINE ANALYSIS OF THE FIXED-BIAS CIRCUIT

We know that the load line is constructed on the output characteristics joining $(0, V_{CC}/R_C)$ and $(V_{CC}, 0)$. The effect of variation in I_B on the Q -point is shown in Fig 2.7.

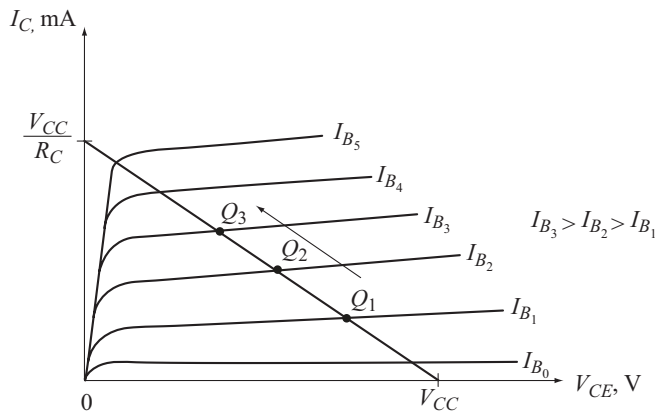


Fig. 2.7 Effect of variation in base current on Q -point

If the value of I_B is increased by decreasing the value of base resistance R_B , the Q -point moves up the load line gradually towards saturation.

Now, if V_{CC} is kept constant and if collector resistance R_C is increased, keeping base current I_B fixed, the Q -point moves to the left into saturation as shown in Fig. 2.8.

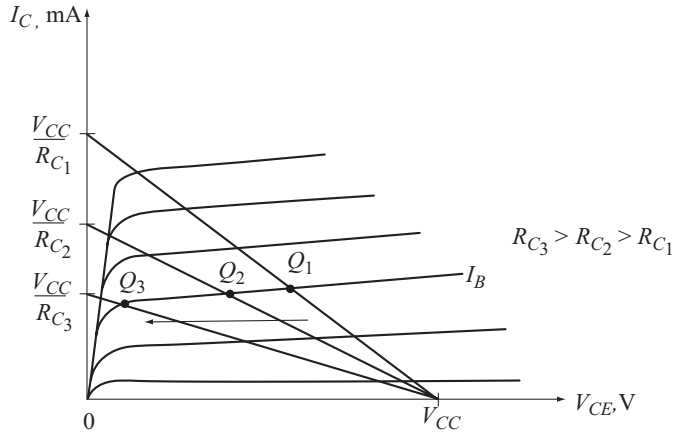


Fig. 2.8 Effect of variation in collector resistance on Q-point

If we now keep R_C fixed while decreasing V_{CC} , the Q-point once again moves towards the left into saturation as shown in Fig. 2.9.

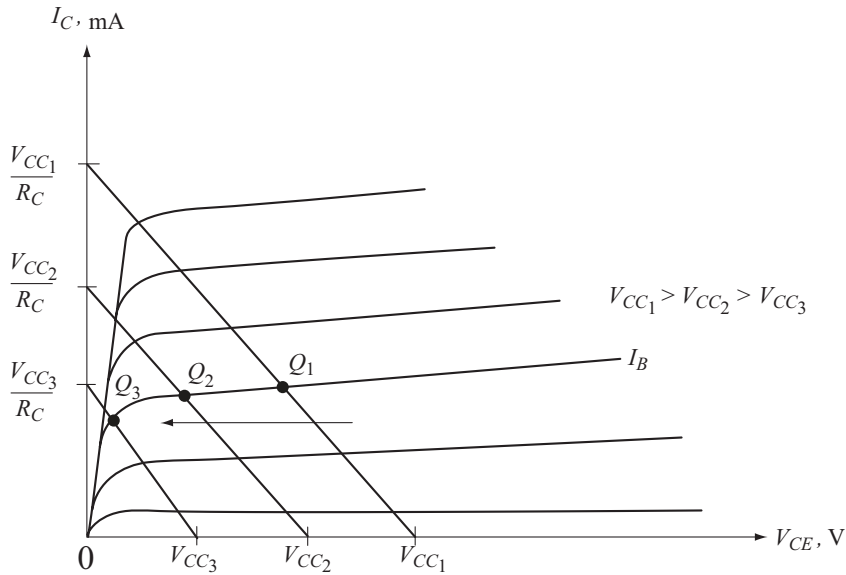


Fig. 2.9 Effect of Variation in V_{CC} on Q-point

◆ 2.7 EMITTER-BIAS CIRCUIT

The addition of an emitter resistance improves the stability of the fixed-bias circuit. Such a configuration known as an emitter-bias or self-bias circuit is shown in Fig. 2.10.

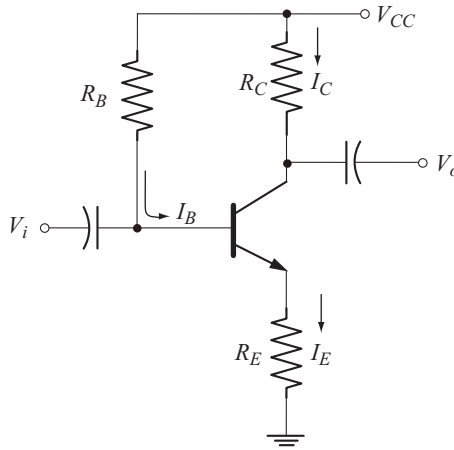


Fig. 2.10 Emitter-bias circuit

Let us proceed with the analysis of this circuit.

Base-emitter Circuit

The base-emitter dc circuit is shown in Fig. 2.11.

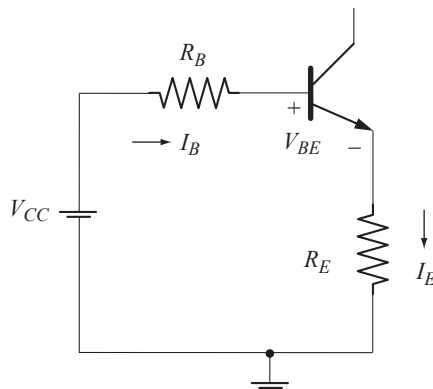


Fig. 2.11 Base-emitter dc circuit of emitter-bias circuit

Applying Kirchoff's voltage law to the base-emitter circuit, we have

$$V_{CC} = I_B R_B + V_{BE} + I_E R_E \quad (2.15)$$

$$I_E = I_C + I_B$$

Using $I_C = \beta I_B$ we have

$$\begin{aligned} I_E &= \beta I_B + I_B \\ I_E &= (\beta + 1) I_B \end{aligned} \quad (2.16)$$

Substituting for I_E in Equation (2.15),

$$\begin{aligned}
 V_{CC} &= I_B R_B + V_{BE} + (\beta + 1) I_B R_E \\
 \therefore V_{CC} - V_{BE} &= I_B [R_B + (\beta + 1) R_E] \\
 \text{or} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \quad (2.17)
 \end{aligned}$$

Stability of the Q-point

Now, let us examine how this circuit stabilizes the operating point. Let us assume for reasons such as increase in temperature, the collector current, I_C increases. This results in an increase in the emitter current, I_E . From Equation (2.15)

$$V_{CC} - V_{BE} = I_B R_B - I_E R_E \quad (2.18)$$

The left hand side of Equation (2.18) is a constant. An increase in I_E increases $I_E R_E$ and to maintain constant $(V_{CC} - V_{BE})$, the term $I_B R_B$ must decrease; R_B being constant, I_B reduces. This in turn reduces I_C ($I_C = \beta I_B$), thereby stabilizing I_C .

With $R_E = 0$, Equation (2.17) reduces to Equation (2.9) as expected. From Equation (2.17) it is clear that the equivalent resistance at the base is

$$R_{\text{Beq}} = R_B + (\beta + 1) R_E \quad (2.19)$$

Thus, the resistance R_E in the emitter circuit gets reflected as $(\beta + 1) R_E$ in the base circuit. Resistance R_E provides a negative feedback to the circuit which minimizes drift in the operating point in comparison with the fixed-bias circuit.

Collector-emitter Circuit

The collector-emitter dc circuit is shown in Fig. 2.12.

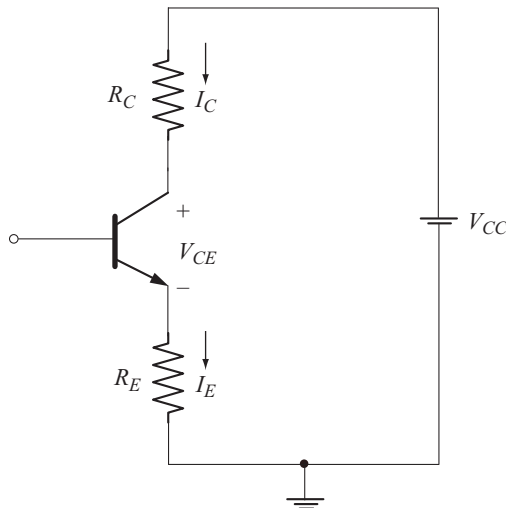


Fig. 2.12 Collector-emitter dc circuit of emitter-bias circuit

Applying Kirchhoff's voltage law, we have,

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E \quad (2.20)$$

Substituting $I_E \simeq I_C$,

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \quad (2.21)$$

Let us now get the nomenclature clear:

Let V_C = Voltage from collector to ground

V_B = Voltage from base to ground and

V_E = Voltage from emitter to ground

$\therefore V_E$ = Voltage across R_E

$$V_E = I_E R_E \quad (2.22)$$

V_C = Collector-emitter voltage + Voltage from emitter to ground

$$V_C = V_{CE} + V_E \quad (2.23)$$

$$= V_{CE} + I_E R_E$$

Substituting for $(V_{CE} + I_E R_E)$ from Equation (2.20)

$$V_C = V_{CC} - I_C R_C \quad (2.24)$$

V_B = Base-emitter voltage + voltage from emitter to ground

$$V_B = V_{BE} + V_E \quad (2.25)$$

$$= V_{BE} + I_E R_E \quad (2.26)$$

Substituting for $(V_{BE} + I_E R_E)$ from Equation (2.15)

$$V_B = V_{CC} - I_B R_B \quad (2.27)$$

Saturation Level

When the transistor is in saturation

$$V_{CE} = V_{CE(\text{sat})} \approx 0 \text{ V}$$

and

$$I_C = I_{C(\text{sat})}$$

Substituting this condition in Equation (2.20) and taking $I_E \approx I_C$ we have

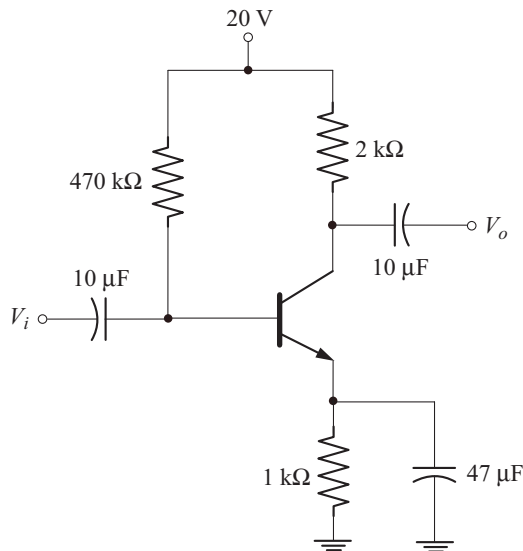
$$V_{CC} = I_{C(\text{sat})} [R_C + R_E]$$

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} \quad (2.28)$$

Example 2.5

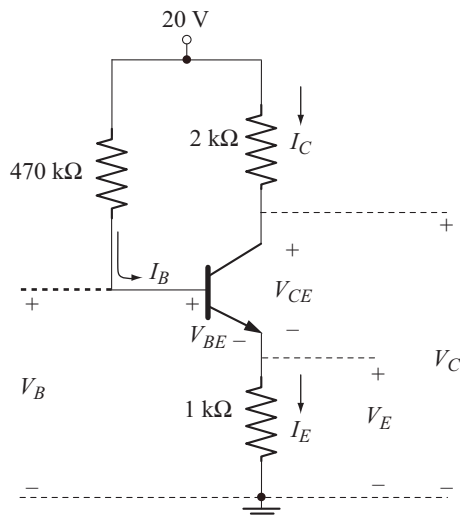
For the emitter-bias circuit shown using silicon transistor with $V_{BE} = 0.7\text{ V}$ and $\beta = 60$, find,

- Base current and collector current
- V_{CE}
- Collector, emitter and base voltages to ground
- V_{BC}
- Estimate I_C and V_{CE} for $\beta = 110$.

**Solution**

- Base current and collector current

For dc analysis we can treat capacitors as open circuit



From Equation (2.17)

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (61 + 1) \text{ k}\Omega} = 36.34 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = 60 \times 36.34 \text{ }\mu\text{A} = 2.18 \text{ mA}$$

(b) V_{CE}

From Equation (2.21)

$$\begin{aligned} V_{CE} &= V_{CC} - I_C (R_C + R_E) \\ &= 20 \text{ V} - 2.18 \text{ mA} [2 \text{ k}\Omega + 1 \text{ k}\Omega] \\ &= 13.46 \text{ V} \end{aligned}$$

(c) Collector, emitter and base voltages to ground

From Equation (2.24)

$$V_C = V_{CC} - I_C R_C = 20 \text{ V} - (2.18 \text{ mA} \times 2 \text{ k}\Omega) = 15.64 \text{ V}$$

From Equation (2.23)

$$V_E = V_C - V_{CE} = 15.64 \text{ V} - 13.46 \text{ V} = 2.18 \text{ V}$$

From Equation (2.25)

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.18 \text{ V} = 2.88 \text{ V}$$

(d) V_{BC}

$$\begin{aligned} V_{BC} &= \text{Voltage from base to collector} \\ &= \text{Voltage from base to ground} - \text{Voltage from collector to ground} \\ &= V_B - V_C = 2.88 \text{ V} - 15.64 \text{ V} = -12.76 \text{ V} \end{aligned}$$

V_{BC} is negative as expected.

(e)

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + [110 + 1] \text{ k}\Omega} = 33.22 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = 110 \times 33.22 \text{ }\mu\text{A} = 3.65 \text{ mA}$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C (R_C + R_E) \\ &= 20 \text{ V} - 3.65 \text{ mA} [2 \text{ k}\Omega + 1 \text{ k}\Omega] \\ &= 9.05 \text{ V} \end{aligned}$$

Let us now tabulate the quiescent currents and voltage for $\beta = 60$ and $\beta = 110$.

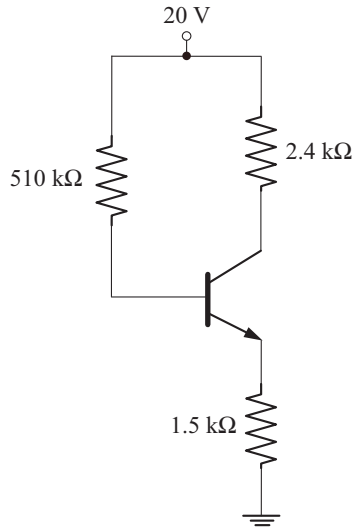
β	I_{BQ}	I_{CQ}	V_{CEQ}
60	36.34 μA	2.18 mA	13.46 V
110	33.22 μA	3.65 mA	9.05 V

Observe that the effect of the negative feedback through the inclusion of the emitter resistance has been to reduce the base current in spite of β being nearly doubled. The operating point remains very much in the active region. Compare this with the results obtained in Example 2.1(e).

Example 2.6

For the emitter-bias circuit shown using silicon transistor with $V_{BE} = 0.7 \text{ V}$ and $\beta = 100$, find

- Quiescent values of base current, collector current and collector to emitter voltage.
- Voltage at collector, base and emitter with respect to ground.



Solution

- (a) I_B , I_C and V_{CE}

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_E}$$

$$= \frac{20 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega + (101)(1.5 \text{ k}\Omega)} = 29.18 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (100)(29.18 \text{ }\mu\text{A}) = 2.92 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C [R_C + R_E]$$

$$= 20 \text{ V} - (2.92 \text{ mA})(2.4 \text{ k}\Omega + 1.5 \text{ k}\Omega) = 8.61 \text{ V}$$

$$V_C = V_{CC} - I_C R_C = 20 \text{ V} - (2.92 \text{ mA})(2.4 \text{ k}\Omega) = 13 \text{ V}$$

$$V_E = I_E R_E = [I_B + I_C] R_E$$

$$= [29.18 \text{ }\mu\text{A} + 2.92 \text{ mA}][1.5 \text{ k}\Omega] = 4.42 \text{ V}$$

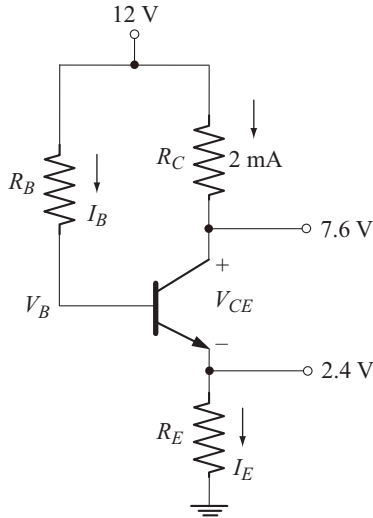
$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 4.42 \text{ V} = 5.12 \text{ V}$$

Example 2.7

For the circuit shown, using silicon transistor with $V_{BE} = 0.7 \text{ V}$ and $\beta = 80$ find

(a) All resistance values

(b) V_{CE} and V_B

**Solution**

Given $V_C = 7.6 \text{ V}$, $V_E = 2.4 \text{ V}$, $I_C = 2 \text{ mA}$

(a) Circuit resistances

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12 \text{ V} - 7.6 \text{ V}}{2 \text{ mA}} = 2.2 \text{ k}\Omega$$

Since, the emitter and the collector currents are approximately equal,

$$I_E \simeq I_C = 2 \text{ mA}$$

$$\therefore R_E = \frac{V_E}{I_E} = \frac{2.4 \text{ V}}{2 \text{ mA}} = 1.2 \text{ k}\Omega$$

$$I_C = \beta I_B$$

$$\therefore I_B = \frac{I_C}{\beta} = \frac{2 \text{ mA}}{80} = 0.025 \text{ mA}$$

Writing KVL equation to the base-emitter circuit,

$$V_{CC} = I_B R_B + V_{BE} + V_E$$

$$\therefore I_B R_B = V_{CC} - V_{BE} - V_E = 12 \text{ V} - 0.7 \text{ V} - 2.4 \text{ V} = 8.9 \text{ V}$$

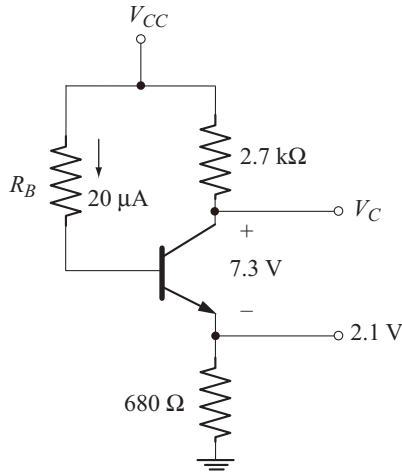
$$\therefore R_B = \frac{8.9 \text{ V}}{I_B} = \frac{8.9 \text{ V}}{0.025 \text{ mA}} = 356 \text{ k}\Omega$$

(b) $V_{CE} = V_C - V_E = 7.6 \text{ V} - 2.4 \text{ V} = 5.2 \text{ V}$
 and $V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.4 \text{ V} = 3.1 \text{ V}$

Example 2.8

The circuit shown below uses silicon transistor with $V_{BE} = 0.7 \text{ V}$. Calculate

- (a) β (b) V_{CC} (c) R_B (d) V_C (e) $I_{C(\text{sat})}$

**Solution**

- (a) β

Given the voltage across the emitter resistance, we can obtain the emitter current as

$$I_E = \frac{V_E}{R_E} = \frac{2.1 \text{ V}}{680 \text{ } \Omega} = 3.09 \text{ mA}$$

The emitter current is approximately equal to collector current.

$$\therefore I_C = 3.09 \text{ mA}$$

$$\text{Now } \beta = \frac{I_C}{I_B} = \frac{3.09 \text{ mA}}{20 \text{ } \mu\text{A}} = 154.5$$

- (b) V_{CC}

From the KVL equation for the collector-emitter circuit, we have

$$\begin{aligned} V_{CC} &= I_C R_C + V_{CE} + V_E \\ &= (3.09 \text{ mA} \times 2.7 \text{ k}\Omega) + 7.3 \text{ V} + 2.1 \text{ V} = 17.74 \text{ V} \end{aligned}$$

- (c) R_B

Writing the KVL equation for the base-emitter circuit

$$V_{CC} = I_B R_B + V_{BE} + V_E$$

$$\begin{aligned} \therefore R_B &= \frac{V_{CC} - V_{BE} - V_E}{I_B} \\ &= \frac{17.74 \text{ V} - 0.7 \text{ V} - 2.1 \text{ V}}{20 \mu\text{A}} = 747 \text{ k}\Omega \end{aligned}$$

(d) V_C

$$V_C = V_{CE} + V_E = 7.3 \text{ V} + 2.1 \text{ V} = 9.4 \text{ V}$$

(e) $I_{C(\text{sat})}$

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} = \frac{17.74 \text{ V}}{2.7 \text{ k}\Omega + 680 \Omega} = 5.248 \text{ mA}$$

◆ 2.8 VOLTAGE DIVIDER BIAS OR UNIVERSAL BIAS CIRCUIT

In both the fixed-bias and the emitter-bias circuits, the quiescent values of I_C and V_{CE} , i.e., the quiescent point is a function of dc current gain β of the transistor. We know that this current gain β is sensitive to temperature and its value keeps varying. A biasing circuit independent or less dependant on β such as the voltage-divider bias circuit is therefore desirable. A voltage divider bias circuit is shown in Fig. 2.13.

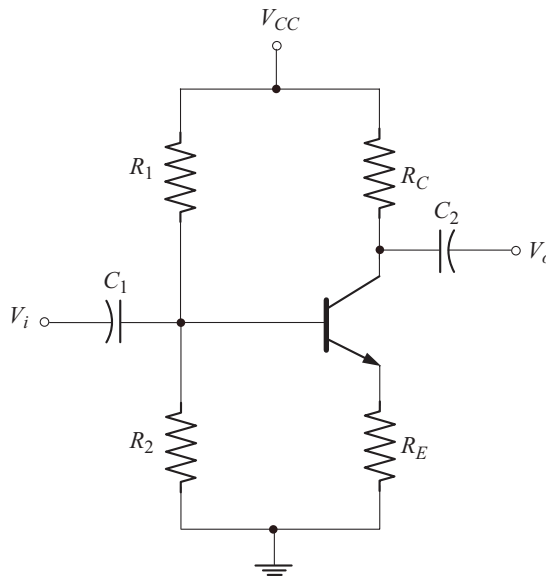


Fig. 2.13 Voltage divider bias or universal bias circuit

The voltage divider bias circuit can be analysed in two methods. First one is the exact method and the second one is the approximate method. Let us analyse the circuit using each of these methods.

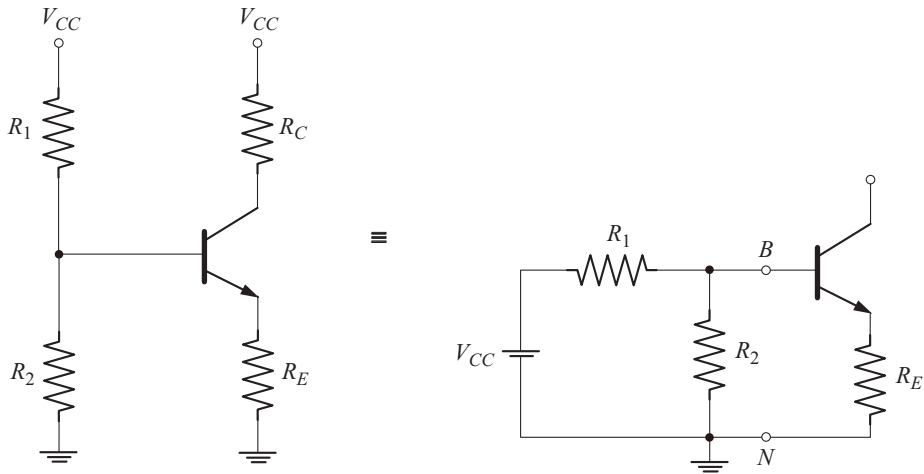


Fig. 2.14 Base circuit of voltage divider bias circuit

2.8.1 Exact Analysis

The input side of the circuit of Fig. 2.13 is redrawn in Fig. 2.14 for the dc analysis.

The Thevenin equivalent of the circuit comprising of V_{CC} , R_1 and R_2 of Fig. 2.14 is drawn in Fig. 2.15(a).

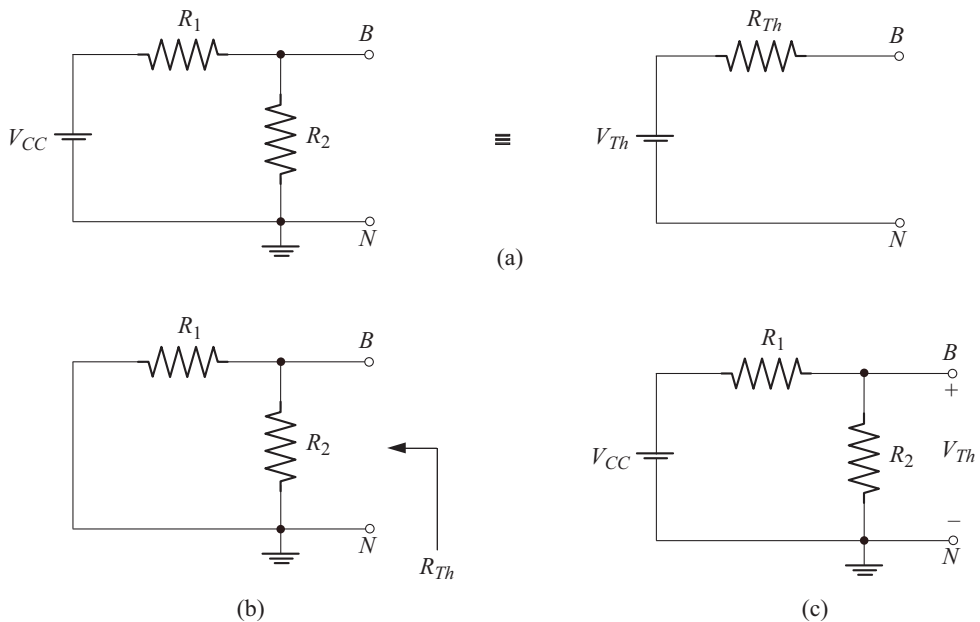


Fig. 2.15 (a) Thevenin equivalent
(b) Determination of R_{Th}
(c) Determination of V_{Th}

The Thevenin equivalent of the circuit to the left of B and N in Fig. 2.14 can be written by computing the Thevenin resistance and Thevenin voltage as seen between B and N .

To find the Thevenin resistance R_{Th} , V_{CC} is reduced to zero in the circuit of Fig. 2.15(a). The resulting circuit is shown in Fig. 2.15(b).

From Fig. 2.15(b), Thevenin resistance is

$$R_{Th} = R_1 \parallel R_2$$

or

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

and from Fig. 2.15(c), Thevenin voltage is

$$V_{Th} = V_{CC} \frac{R_2}{R_1 + R_2}$$

The circuit of Fig. 2.14 is redrawn in Fig. 2.16 after substituting the Thevenin equivalent between B and N .

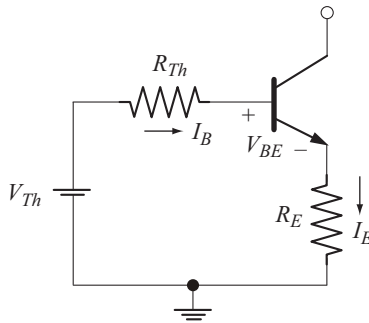


Fig. 2.16 Base circuit with Thevenin equivalent

Applying Kirchhoff's voltage law to the base-emitter circuit in Fig. 2.16, we get

$$V_{Th} = R_{Th} I_B + V_{BE} + I_E R_E \quad (2.29)$$

Now

$$I_C + I_B = I_E \quad (2.30)$$

and

$$I_C = \beta I_B$$

Substituting in Equation (2.30)

$$\beta I_B + I_B = I_E$$

or

$$I_E = (\beta + 1) I_B \quad (2.31)$$

Equation (2.29) now becomes

$$V_{Th} = R_{Th} I_B + V_{BE} + (\beta + 1) I_B R_E$$

$$V_{Th} - V_{BE} = [R_{Th} + (\beta + 1) R_E] I_B$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1) R_E} \quad (2.32)$$

The collector circuit of Fig. 2.13 is shown in Fig 2.17.

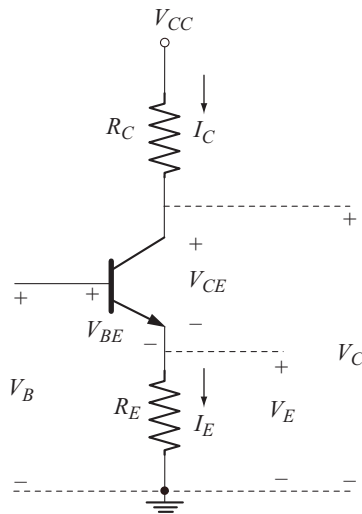


Fig. 2.17 Collector circuit of voltage divider bias

Applying KVL to the collector circuit of Fig. 2.17,

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E \quad (2.33)$$

Using

$$I_C \approx I_E \quad (2.34)$$

Equation (2.33) thus become

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \quad (2.35)$$

Further from Fig. 2.17,

$$\begin{aligned} V_E &= \text{Voltage across } R_E \\ V_E &= I_E R_E \end{aligned} \quad (2.36)$$

$$\begin{aligned} V_C &= \text{Collector to ground voltage} \\ V_C &= V_{CE} + V_E \end{aligned} \quad (2.37)$$

$$\begin{aligned} V_B &= \text{Base to ground voltage} \\ V_B &= V_{BE} + V_E \end{aligned} \quad (2.38)$$

In this circuit also negative feedback is provided through the emitter resistance. As a result I_C and V_{CE} become almost independent of β . For this reason voltage divider bias is also called as beta independent circuit.

Transistor Saturation

When the transistor is in saturation

$$\begin{aligned} V_{CE} &= V_{CE(\text{sat})} \approx 0 \text{ V} \\ \text{and} \quad I_C &= I_{C(\text{sat})} \end{aligned}$$

Substituting this condition in Equation (2.33) and taking $I_E \approx I_C$, we get

$$\begin{aligned} V_{CC} &= I_{C(\text{sat})} [R_C + R_E] \\ I_{C(\text{sat})} &= \frac{V_{CC}}{R_C + R_E} \end{aligned} \quad (2.39)$$

Note that Equation (2.39) is same as that obtained for emitter bias circuit in Equation (2.28).

2.8.2 Approximate Analysis

We know that, the resistance R_E in the emitter circuit gets reflected as $(1 + \beta) R_E$ in the base circuit. Therefore the circuit between base and ground of Fig. 2.14 can be replaced by an equivalent resistance. $R_i = (1 + \beta) R_E$ as shown in Fig. 2.18.

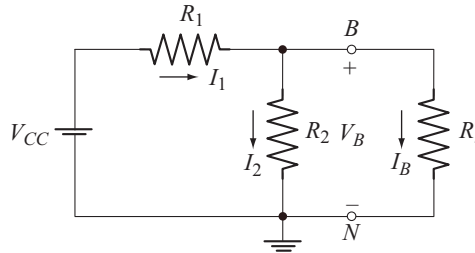


Fig. 2.18 Input circuit for approximate analysis

For the circuit of Fig. 2.18 using KCL we can write

$$I_1 = I_2 + I_B \quad (2.40)$$

$$R_i = (1 + \beta) R_E \approx \beta R_E, \quad \text{since } \beta \gg 1$$

$$I_B = \frac{V_B}{R_i} \quad \text{and} \quad I_2 = \frac{V_B}{R_2}$$

$$\text{if} \quad R_i = \beta R_E \geq 10 R_2 \quad (2.41)$$

then $I_B \leq 0.1 I_2$, hence I_B can be neglected in Equation (2.40). As a result we get

$$I_1 \approx I_2$$

Applying KVL to the circuit of Fig. 2.18 we get

$$V_{CC} = I_1 R_1 + I_2 R_2 = I_2 [R_1 + R_2]$$

$$\therefore \quad I_2 = \frac{V_{CC}}{R_1 + R_2}$$

$$\text{Now} \quad V_B = I_2 R_2$$

$$\text{or} \quad V_B = \frac{V_{CC} R_2}{R_1 + R_2} \quad (2.42)$$

Observe that expression for V_B is identical to that of V_{Th}

$$\begin{aligned} \text{But} \quad & V_B = V_E + V_{BE} \\ \therefore \quad & V_E = V_B - V_{BE} \end{aligned} \quad (2.43)$$

The emitter current I_E is given by

$$I_E = \frac{V_E}{R_E} \quad (2.44)$$

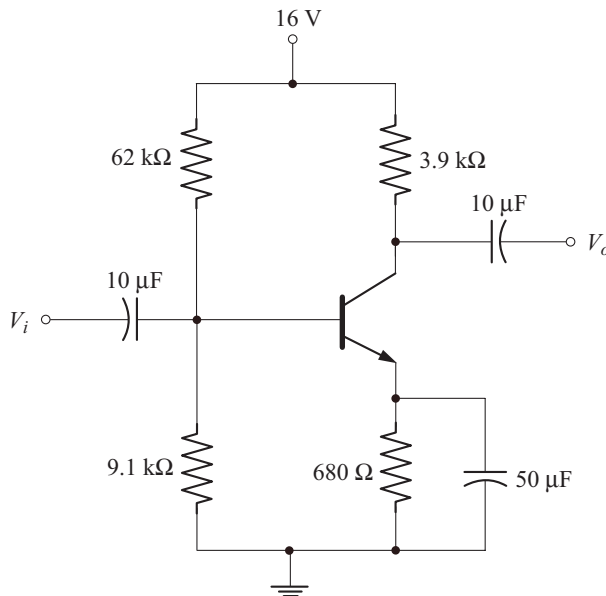
$$\text{and} \quad I_C \approx I_E \quad (2.45)$$

From the KVL equation of collector - emitter circuit

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\ \text{Using} \quad & I_C \approx I_E \\ V_{CE} &= V_{CC} - I_C [R_C + R_E] \end{aligned} \quad (2.46)$$

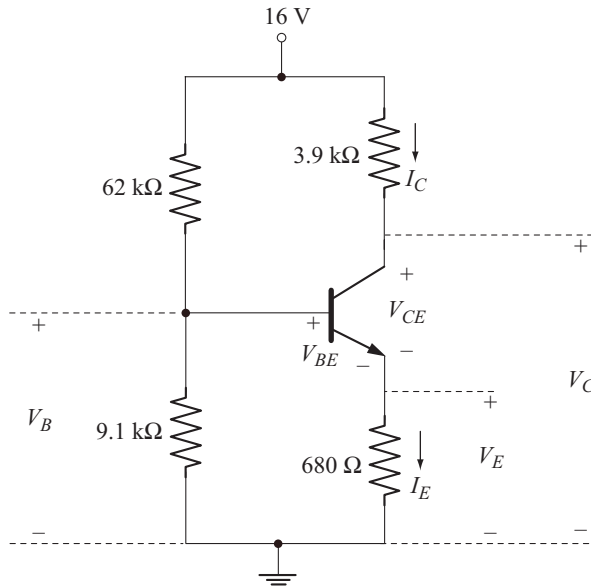
Example 2.9

- Find the quiescent base current, collector current and V_{CE} for the circuit shown using silicon transistor with $V_{BE} = 0.7$ V and $\beta = 80$.
- Determine the values of collector, emitter and base voltages with respect to ground.
- Repeat (a) for $\beta = 150$.
- Draw the dc load line and locate the Q points corresponding to $\beta = 80$ and $\beta = 150$.



Solution

Let us draw the dc circuit with all voltages and currents marked.



(a)

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{62 \text{ k}\Omega \times 9.1 \text{ k}\Omega}{62 \text{ k}\Omega + 9.1 \text{ k}\Omega} = 7.94 \text{ k}\Omega$$

$$V_{Th} = V_{CC} \frac{R_2}{R_1 + R_2} = 16 \text{ V} \times \frac{9.1 \text{ k}\Omega}{62 \text{ k}\Omega + 9.1 \text{ k}\Omega} = 2.05 \text{ V}$$

The base current can now be obtained from

$$\begin{aligned} I_{BQ} &= \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\ &= \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (80 + 1)0.68 \text{ k}\Omega} = 21.42 \text{ }\mu\text{A} \end{aligned}$$

$$I_{CQ} = \beta I_B = 80 \times 21.42 \text{ }\mu\text{A} = 1.71 \text{ mA}$$

$$\begin{aligned} V_{CEQ} &= V_{CC} - I_C (R_C + R_E) \\ &= 16 \text{ V} - 1.71 \text{ mA} [3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega] = 8.17 \text{ V} \end{aligned}$$

(b) Collector to ground voltage is

$$V_C = V_{CC} - I_C R_C = 16 \text{ V} - (1.71 \text{ mA} \times 3.9 \text{ k}\Omega) = 9.33 \text{ V}$$

Emitter to ground voltage is

$$V_E = I_E R_E \simeq I_C R_E = 1.71 \text{ mA} \times 0.68 \text{ k}\Omega = 1.16 \text{ V}$$

and the base to ground voltage is

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 1.16 \text{ V} = 1.86 \text{ V}$$

(c) Now for $\beta = 150$

$$I_{BQ} = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

$$= \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (150 + 1)0.68 \text{ k}\Omega} = 12.2 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = 150 \times 12.2 \mu\text{A} = 1.83 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_C (R_C + R_E)$$

$$= 16 \text{ V} - 1.83 \text{ mA} (3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega) = 7.62 \text{ V}$$

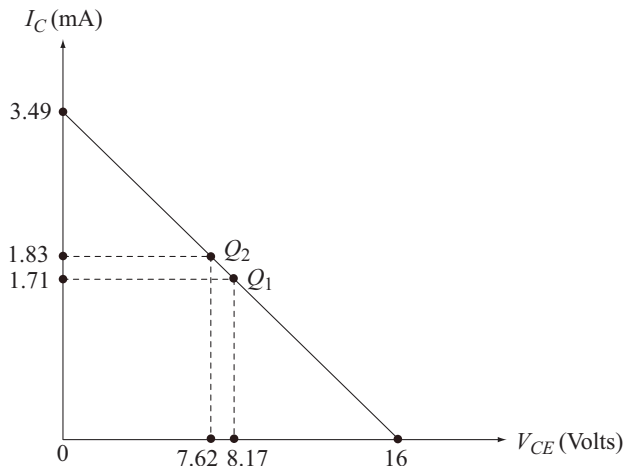
Let us now tabulate the quiescent currents and voltage for $\beta = 80$ and $\beta = 150$.

β	I_{BQ}	I_{CQ}	V_{CEQ}
80	21.42 μA	1.71 mA	8.17 V
150	12.2 μA	1.83 mA	7.62 V

Observe that even on nearly doubling current gain β , I_{BQ} reduces by almost half while I_{CQ} and V_{CEQ} marginally change indicating good stability of the operating point. Compare this with the results obtained in Examples 2.1 and 2.6. In the case of the voltage divider bias circuit, the base current drastically reduces to keep the increase in collector current, due to increase in β at a minimum.

(d) DC load line

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} = \frac{16 \text{ V}}{3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega} = 3.49 \text{ mA}$$

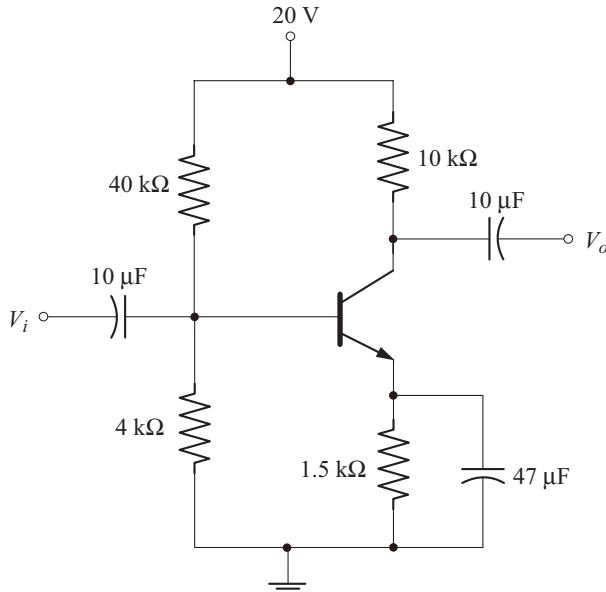


Example 2.10

For the voltage divider bias configuration shown below.

- Find I_C and V_{CE} using exact analysis.
- Find I_C and V_{CE} using approximate analysis.
- Find $I_{C(sat)}$.
- Compare the results obtained (a) and (b) and comment.

Assume silicon transistor with $\beta = 150$.

**Solution****(a) Exact Analysis**

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{(40 \text{ k}\Omega)(4 \text{ k}\Omega)}{40 \text{ k}\Omega + 4 \text{ k}\Omega} = 3.63 \text{ k}\Omega$$

$$V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(20 \text{ V})(4 \text{ k}\Omega)}{40 \text{ k}\Omega + 4 \text{ k}\Omega} = 1.82 \text{ V}$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (1 + \beta) R_E}$$

$$= \frac{1.82 \text{ V} - 0.7 \text{ V}}{3.63 \text{ k}\Omega + (151)(1.5 \text{ k}\Omega)} = 4.86 \mu\text{A}$$

$$I_C = \beta I_B = (150)(4.86 \mu\text{A}) = 0.729 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C [R_C + R_E]$$

$$= 20 \text{ V} - 0.729 \text{ mA} [10 \text{ k}\Omega + 1.5 \text{ k}\Omega] = 11.62 \text{ V}$$

(b) *Approximate Analysis*

$$\beta R_E = (150)(1.5 \text{ k}\Omega) = 225 \text{ k}\Omega$$

$$10 R_2 = (10)(4 \text{ k}\Omega) = 40 \text{ k}\Omega$$

Note that $\beta R_E > 10 R_2$

Hence we can use approximate analysis

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} = 1.82 \text{ V} \quad (\text{same as } V_{Th})$$

$$V_E = V_B - V_{BE} = 1.82 \text{ V} - 0.7 \text{ V} = 1.12 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.12 \text{ V}}{1.5 \text{ k}\Omega} = 0.746 \text{ mA}$$

$$I_C \approx I_E = 0.746 \text{ mA}$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C [R_C + R_E] \\ &= 20 \text{ V} - 0.746 \text{ mA} [10 \text{ k}\Omega + 1.5 \text{ k}\Omega] = 11.42 \text{ V} \end{aligned}$$

$$(c) \quad I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} = \frac{20 \text{ V}}{10 \text{ k}\Omega + 1.5 \text{ k}\Omega} = 1.74 \text{ mA}$$

(d) The results of exact and approximate analysis are compared in the following table

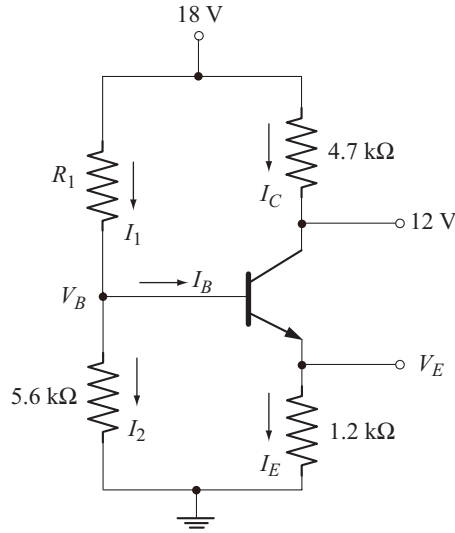
<i>Parameter</i>	<i>Exact analysis</i>	<i>Approximate analysis</i>
I_C	0.729 mA	0.746 mA
V_{CE}	11.62 V	11.42 V

Observe that there is only a slight difference in the values since the condition $\beta R_E \geq 10 R_E$ is satisfied.

Example 2.11

For the circuit shown in the figure below using silicon transistor with $V_{BE} = 0.7 \text{ V}$. Find

- Collector current
- Emitter and base voltages with respect to ground
- Value of resistance R_1 .

**Solution**

(a)

$$V_{CC} = I_C R_C + V_C$$

 $\therefore$

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18 \text{ V} - 12 \text{ V}}{4.7 \text{ k}\Omega} = 1.27 \text{ mA}$$

(b)

$$V_E = I_E R_E \approx I_C R_E = (1.27 \text{ mA})(1.2 \text{ k}\Omega) = 1.52 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 1.52 \text{ V} = 2.22 \text{ V}$$

(c)

$$V_{CC} = I_1 R_1 + I_2 R_2 \quad (\text{A})$$

KCL at node V_B yields

$$I_1 = I_B + I_2$$

Neglecting I_B , we get

$$I_1 \approx I_2$$

Now from Equation (A) we have

$$R_1 + R_2 = \frac{V_{CC}}{I_2} \quad (\text{B})$$

But $V_B = I_2 R_2$ $\therefore$

$$I_2 = \frac{V_B}{R_2} = \frac{2.22 \text{ V}}{5.6 \text{ k}\Omega} = 0.39 \text{ mA}$$

From Equation (B) we have

$$R_1 + R_2 = \frac{18 \text{ V}}{0.39 \text{ mA}} = 46.15 \text{ k}\Omega$$

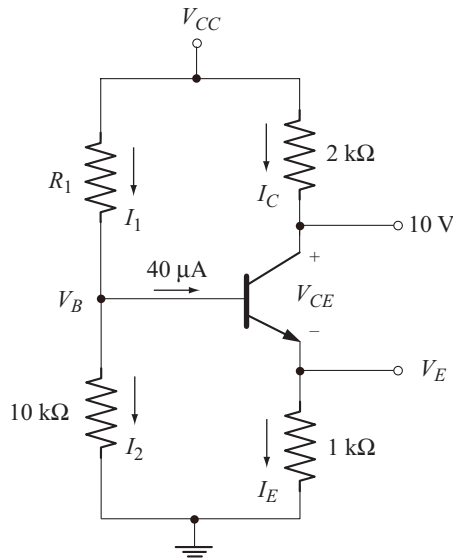
$$R_1 = 46.15 \text{ k}\Omega - R_2 = 46.15 \text{ k}\Omega - 5.6 \text{ k}\Omega = 40.55 \text{ k}\Omega$$

Example 2.12

For the voltage divider bias configuration shown below determine

- (a) I_C and V_E
- (b) V_{CC} and V_{CE}
- (c) V_B and R_1

Assume silicon transistor with $\beta = 80$.

**Solution**

- (a) I_C and V_{CE}

$$I_C = \beta I_B = (80)(40 \mu\text{A}) = 3.2 \text{ mA}$$

$$I_E = I_B + I_C = 40 \mu\text{A} + 3.2 \text{ mA} = 3.24 \text{ mA}$$

$$V_E = I_E R_E = (3.24 \text{ mA})(1 \text{ k}\Omega) = 3.24 \text{ V}$$

- (b) V_{CC} and V_{CE}

$$V_{CC} = I_C R_C + V_C = (3.2 \text{ mA})(2 \text{ k}\Omega) + 10 \text{ V} = 16.4 \text{ V}$$

$$V_{CE} = V_C - V_E = 10 \text{ V} - 3.2 \text{ V} = 6.76 \text{ V}$$

- (c) V_B and R_1

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 3.24 \text{ V} = 3.94 \text{ V}$$

$$V_{CC} = I_1 R_1 + V_B$$

$$R_1 = \frac{V_{CC} - V_B}{I_1}$$

(A)

$$I_1 = I_B + I_2$$

$$I_2 = \frac{V_B}{R_2} = \frac{3.94 \text{ V}}{10 \text{ k}\Omega} = 0.394 \text{ mA}$$

Now

$$I_1 = 40 \mu\text{A} + 0.394 \text{ mA} = 0.434 \text{ mA}$$

From Equation (A)

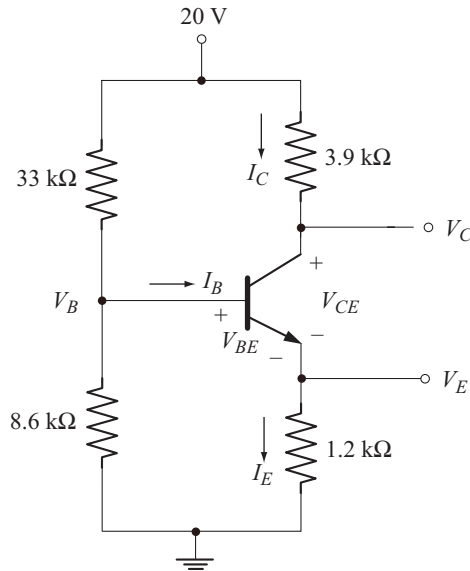
$$R_1 = \frac{16.4 \text{ V} - 3.94 \text{ V}}{0.434 \text{ mA}} = 28.7 \text{ k}\Omega$$

Example 2.13

For the voltage-divider configuration shown below, using the approximate analysis calculate

- (a) V_B
- (b) I_B and I_C
- (c) V_E and V_{CE}

Assume silicon transistor with $\beta = 110$.



Solution

$$\beta R_E = (110)(1.2 \text{ k}\Omega) = 132 \text{ k}\Omega$$

$$10 R_2 = (10)(8.6 \text{ k}\Omega) = 86 \text{ k}\Omega$$

Since $\beta R_E > 10 R_2$, we can use approximate analysis

- (a) V_B

$$V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(20 \text{ V})(8.6 \text{ k}\Omega)}{33 \text{ k}\Omega + 8.6 \text{ k}\Omega} = 4.13 \text{ V}$$

(b) I_B and I_C

$$V_B = V_{BE} + I_E R_E = V_{BE} + [1 + \beta] I_B R_E$$

$$I_B = \frac{V_B - V_{BE}}{[1 + \beta] R_E} = \frac{4.13 \text{ V} - 0.7 \text{ V}}{(111) (1.2 \text{ k}\Omega)} = 25.75 \mu\text{A}$$

$$I_C = \beta I_B = (110) (25.75 \mu\text{A}) = 2.83 \text{ mA}$$

(c) V_E and V_{CE}

$$V_E = I_E R_E$$

$$I_E = I_B + I_C = 25.75 \mu\text{A} + 2.83 \text{ mA} = 2.85 \text{ mA}$$

$$V_E = (2.85 \text{ mA}) (1.2 \text{ k}\Omega) = 3.42 \text{ V}$$

$$V_{CC} = I_C R_C + V_{CE} + V_E$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - V_E \\ &= 20 \text{ V} - (2.83 \text{ mA}) (3.9 \text{ k}\Omega) - 3.42 \text{ V} = 5.54 \text{ V} \end{aligned}$$

◆ 2.9 COLLECTOR FEEDBACK BIAS

The collector feedback bias circuit is realized by introducing feedback path from the collector to base as shown in Fig. 2.19. The operating point of this circuit is less sensitive to variations in temperature and β when compared to the fixed-bias and emitter-bias circuits.

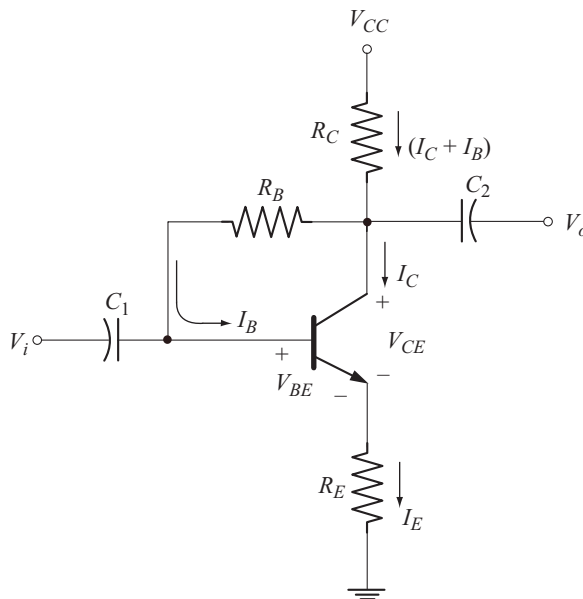


Fig. 2.19 Collector feedback bias circuit

Expressions for I_B and I_C

The dc circuit is shown in Fig. 2.20.

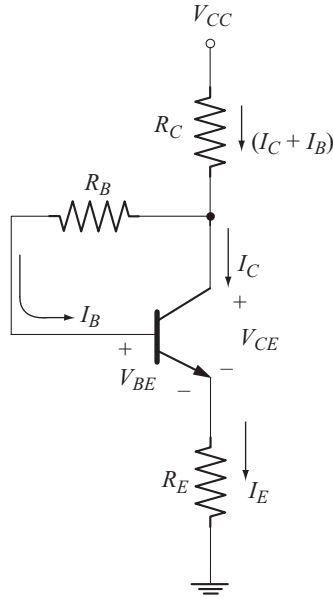


Fig. 2.20 DC circuit

Writing the Kirchhoff's voltage law equation for the base-emitter circuit, we get

$$V_{CC} = (I_C + I_B) R_C + I_B R_B + V_{BE} + I_E R_E \quad (2.47)$$

Now
and

$$I_C + I_B \simeq I_C \quad \text{since } I_B \ll I_C$$

$$I_C \simeq I_E$$

$\therefore$ Equation (2.47) becomes

$$V_{CC} = I_C R_C + I_B R_B + V_{BE} + I_C R_E$$

Substituting $I_C = \beta I_B$

$$V_{CC} = \beta I_B R_C + I_B R_B + V_{BE} + \beta I_B R_E$$

$$V_{CC} = I_B R_B + \beta (R_C + R_E) I_B + V_{BE}$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta (R_C + R_E)} \quad (2.48)$$

On comparison with Equation (2.17), it is clear that in the collector feedback circuit besides R_E , collector resistance R_C also gets reflected in the base circuit.

Collector current,

$$I_C = \beta I_B$$

$$\therefore I_C = \frac{\beta (V_{CC} - V_{BE})}{R_B + \beta (R_C + R_E)} \quad (2.49)$$

If β is large, then $\beta [R_C + R_E] \gg R_B$

$$\therefore R_B + \beta [R_C + R_E] \approx \beta [R_C + R_E]$$

Now Equation (2.49) reduces to

$$I_C \approx \frac{V_{CC} - V_{BE}}{R_C + R_E} \quad (2.50)$$

Observe that the collector current becomes independent of β under these conditions and hence is independent of variations in β .

Expression for V_{CE}

Applying KVL to the collector-emitter circuit of Fig. 2.20 we get

$$V_{CC} = R_C [I_C + I_B] + V_{CE} + I_E R_E \quad (2.51)$$

As in the computation of base current,

$$\begin{aligned} I_C + I_B &\simeq I_C \quad \text{and} \quad I_E \simeq I_C \\ \therefore V_{CC} &= I_C R_C + V_{CE} + I_C R_E \\ \therefore V_{CE} &= V_{CC} - I_C (R_C + R_E) \end{aligned} \quad (2.52)$$

which is the same as Equation (2.35) derived for the voltage-divider bias circuit.

Transistor Saturation

When the transistor is in saturation, $V_{CE(\text{sat})} \approx 0$. From Equation (2.52), the collector current at saturation is given by

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} \quad (2.53)$$

Observe that $I_{C(\text{sat})}$ is same as that of emitter bias and voltage divider bias circuits given in Equations (2.28) and (2.39) respectively.

Example 2.14

The following circuit values are given for the collector feedback bias circuit of Fig. 2.19.

$V_{CC} = 12 \text{ V}$ $R_C = 4.7 \text{ k}\Omega$ $R_B = 220 \text{ k}\Omega$ $R_E = 1 \text{ k}\Omega$. Assuming silicon transistor

- Find the operating point for the circuit shown using silicon transistor with $\beta = 90$.
- Find the operating point for $\beta = 150$.

Solution

- When $\beta = 90$

$$\begin{aligned} I_B &= \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega + 90(4.7 \text{ k}\Omega + 1 \text{ k}\Omega)} = 15.42 \text{ }\mu\text{A} \\ I_C &= \beta I_B = 90 \times 15.42 \text{ }\mu\text{A} = 1.39 \text{ mA} \end{aligned}$$

$$\begin{aligned}
 V_{CE} &= V_{CC} - I_C (R_C + R_E) \\
 &= 12 \text{ V} - 1.39 \text{ mA} \times (4.7 \text{ k}\Omega + 1 \text{ k}\Omega) = 4.08 \text{ V}
 \end{aligned}$$

(b) For $\beta = 150$

$$\begin{aligned}
 I_B &= \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega + 150(4.7 \text{ k}\Omega + 1 \text{ k}\Omega)} = 10.51 \text{ }\mu\text{A} \\
 I_C &= \beta I_B = 150 \times 10.51 \text{ }\mu\text{A} = 1.58 \text{ mA} \\
 V_{CE} &= V_{CC} - I_C (R_C + R_E) \\
 &= 12 \text{ V} - 1.58 \text{ mA} (4.7 \text{ k}\Omega + 1 \text{ k}\Omega) = 3 \text{ V}
 \end{aligned}$$

Let us now tabulate the results for $\beta = 90$ and $\beta = 150$.

β	I_{BQ}	I_{CQ}	V_{CEQ}
90	15.42 μA	1.39 mA	4.08 V
150	10.51 μA	1.58 mA	3V

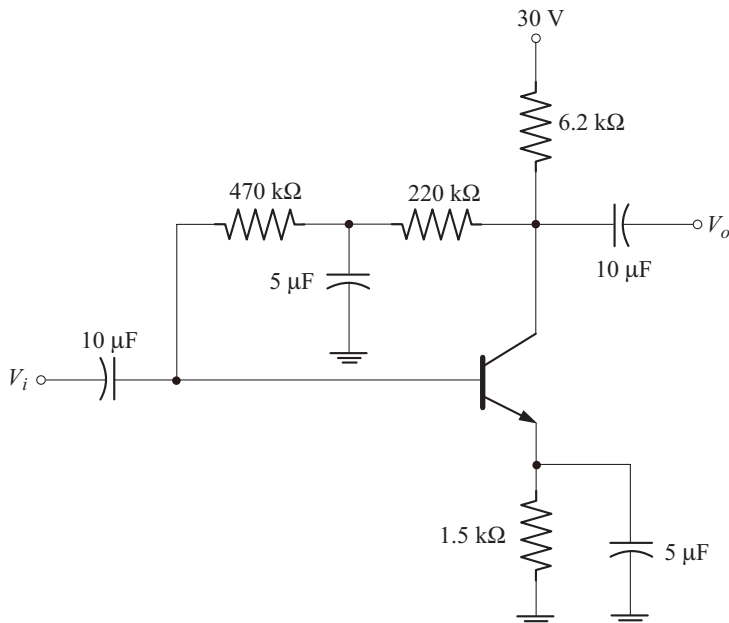
Observe that a large change in β causes a small shift in the operating point well within the active region.

Example 2.15

For the circuit shown below, determine

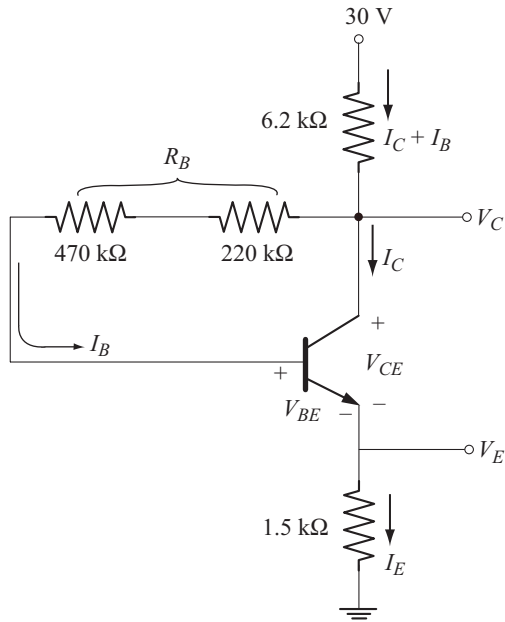
- (a) I_B and I_C (b) V_{CE} and V_C (c) V_E and $I_{C(\text{sat})}$

Assume silicon transistor with $\beta = 100$.



Solution

Let us write the dc circuit by open circuiting all capacitors.



From the circuit

$$R_B = 470 \text{ k}\Omega + 220 \text{ k}\Omega = 690 \text{ k}\Omega$$

(a) I_B and I_C

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{30 \text{ V} - 0.7 \text{ V}}{690 \text{ k}\Omega + 100[6.2 \text{ k}\Omega + 1.5 \text{ k}\Omega]} = 20.07 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (100)(20.07 \text{ }\mu\text{A}) = 2.01 \text{ mA}$$

(b) V_{CE} and V_C

$$\begin{aligned} V_{CE} &= V_{CC} - I_C [R_C + R_E] \quad [\text{Neglecting } I_B] \\ &= 30 \text{ V} - 2.01 \text{ mA} [6.2 \text{ k}\Omega + 1.5 \text{ k}\Omega] = 14.52 \text{ V} \end{aligned}$$

$$V_C = V_{CC} - I_C R_C = 30 \text{ V} - (2.01 \text{ mA})(6.2 \text{ k}\Omega) = 17.54 \text{ V}$$

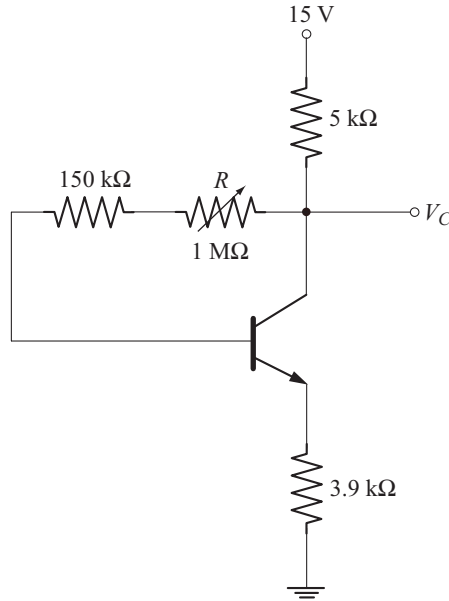
(c) V_E and $I_{C(\text{sat})}$

$$V_E = I_E R_E \approx I_C R_E = (2.01 \text{ mA})(1.5 \text{ k}\Omega) = 3.02 \text{ V}$$

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} = \frac{30 \text{ V}}{6.2 \text{ k}\Omega + 1.5 \text{ k}\Omega} = 3.89 \text{ mA}$$

Example 2.16

For the circuit shown determine the range of possible values for V_C . Assume silicon transistor with $\beta = 200$.

**Solution:**

Let $R_B = 150 \text{ k}\Omega + R$

R varies between 0 and 1 M Ω .

When $R = 0 \Omega$, $R_B = 150 \text{ k}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{15 \text{ V} - 0.7 \text{ V}}{150 \text{ k}\Omega + 200(5 \text{ k}\Omega + 3.9 \text{ k}\Omega)} = 7.4 \mu\text{A}$$

$$I_C = \beta I_B = (200)(7.4 \mu\text{A}) = 1.48 \text{ mA}$$

$$\begin{aligned} V_C &= V_{CC} - [I_C + I_B] R_C \\ &= 15 \text{ V} - [1.48 \text{ mA} + 7.4 \mu\text{A}] 5 \text{ k}\Omega = 7.56 \text{ V} \end{aligned}$$

When $R = 1 \text{ M}\Omega$, $R_B = 150 \text{ k}\Omega + 1 \text{ M}\Omega = 1.15 \text{ M}\Omega$

$$I_B = \frac{15 \text{ V} - 0.7 \text{ V}}{1.15 \text{ M}\Omega + 200(5 \text{ k}\Omega + 3.9 \text{ k}\Omega)} = 4.88 \mu\text{A}$$

$$I_C = \beta I_B = (200)(4.88 \mu\text{A}) = 0.976 \text{ mA}$$

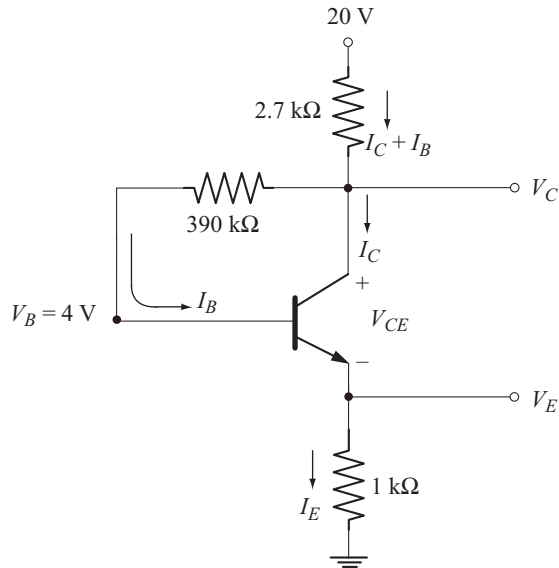
$$V_C = 15 \text{ V} - [0.976 \text{ mA} + 4.88 \mu\text{A}] 5 \text{ k}\Omega = 10 \text{ V}$$

V_C changes from 7.56 V to 10 V when R is varied from 0 to 1 M Ω .

Example 2.17

For the circuit shown below determine

- (a) V_E and I_E
- (b) V_C and I_B
- (c) I_C , V_{CE} and β

**Solution**

- (a) V_E and I_E

$$V_B = V_{BE} + V_E$$

$$V_E = V_B - V_{BE} = 4\text{ V} - 0.7\text{ V} = 3.3\text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{3.3\text{ V}}{1\text{ k}\Omega} = 3.3\text{ mA}$$

- (b) V_C and I_B

$$V_{CC} = (I_C + I_B) R_C + V_C$$

$$V_C = V_{CC} - I_E R_C = 20\text{ V} - (3.3\text{ mA})(2.7\text{ k}\Omega) = 11.09\text{ V}$$

$$I_B = \frac{V_C - V_B}{R_B} = \frac{11.09\text{ V} - 4\text{ V}}{390\text{ k}\Omega} = 18.17\text{ }\mu\text{A}$$

- (c) I_C , V_{CE} and β

$$I_C = I_E - I_B = 3.3\text{ mA} - 18.17\text{ }\mu\text{A} = 3.28\text{ mA}$$

$$V_{CE} = V_C - V_E = 11.09\text{ V} - 3.3\text{ V} = 7.79\text{ V}$$

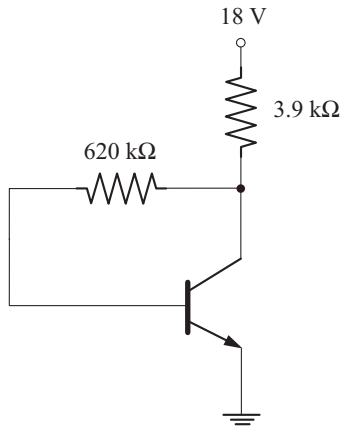
$$\beta = \frac{I_C}{I_B} = \frac{3.28\text{ mA}}{18.17\text{ }\mu\text{A}} = 180.5$$

Example 2.18

For the circuit shown below calculate

- (a) I_{CQ} and V_{CEQ}
- (b) V_E and V_B
- (c) V_C and V_{BC}

Assume silicon transistor with $\beta = 150$.

**Solution**

The given circuit is nothing but collector feedback circuit with $R_E = 0 \Omega$.

- (a) I_{CQ} and V_{CEQ}

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta[R_C + R_E]} = \frac{18 \text{ V} - 0.7 \text{ V}}{620 \text{ k}\Omega + (150)(3.9 \text{ k}\Omega)} = 14.35 \mu\text{A}$$

$$I_C = \beta I_B = (150)(14.35 \mu\text{A}) = 2.15 \text{ mA}$$

$$\begin{aligned} V_{CE} &= V_{CC} - [I_C + I_B] R_C \\ &= 18 \text{ V} - [2.15 \text{ mA} + 14.35 \mu\text{A}] [3.9 \text{ k}\Omega] = 9.55 \text{ V} \end{aligned}$$

- (b) V_E and V_B

$$V_E = I_E R_E = (I_E)(0 \Omega) = 0 \text{ V}$$

$$V_B = V_{BE} + V_E = V_{BE} = 0.7 \text{ V}$$

- (c) V_C and V_{BC}

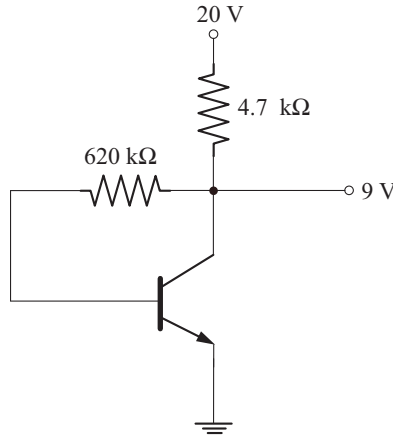
$$V_C = V_{CE} + V_E = V_{CE} = 9.55 \text{ V}$$

$$V_{BC} = V_B - V_C = 0.7 \text{ V} - 9.55 \text{ V} = -8.85 \text{ V}$$

Example 2.19

For the circuit shown below calculate

- (a) I_B and I_C
- (b) β and V_{CE}

**Solution**

Given $V_C = 9 \text{ V}$

Since $R_E = 0 \Omega$

$$V_E = 0 \text{ V}$$

$\therefore$

$$V_C = V_{CE} = 9 \text{ V}$$

and

$$V_B = V_{BE} = 0.7 \text{ V}$$

(a) I_B and I_C

$$I_B = \frac{V_C - V_B}{R_B} = \frac{9 \text{ V} - 0.7 \text{ V}}{620 \text{ k}\Omega} = 13.38 \mu\text{A}$$

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{20 \text{ V} - 9 \text{ V}}{4.7 \text{ k}\Omega} = 2.34 \text{ mA}$$

(b) β and V_{CE}

$$\beta = \frac{I_C}{I_B} = \frac{2.34 \text{ mA}}{13.38 \mu\text{A}} = 174.88$$

$$V_{CE} = V_C = 9 \text{ V}$$

Example 2.20

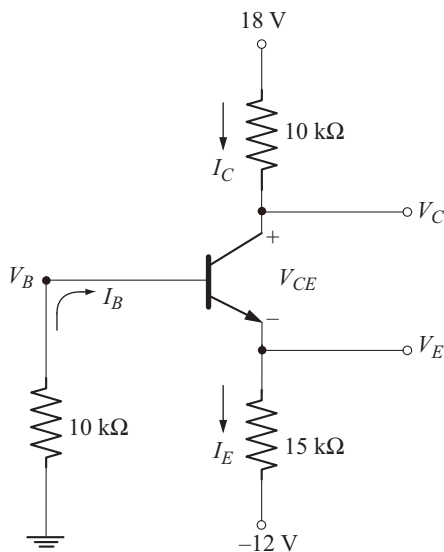
For the circuit shown below determine

(a) I_B and I_C

(b) V_E and V_B

(c) V_C and V_{CE}

Assume silicon transistor with $\beta = 150$.

**Solution**(a) I_B and I_C

Applying KVL to base-emitter circuit, we get

$$-I_B (10 \text{ k}\Omega) - V_{BE} - I_E (15 \text{ k}\Omega) + 12 \text{ V} = 0 \quad (\text{A})$$

$$I_E = (1 + \beta) I_B = 151 I_B$$

Using this relation in Equation (A) we have

$$12 \text{ V} - V_{BE} = I_B (10 \text{ k}\Omega) + 151 I_B (15 \text{ k}\Omega)$$

$$I_B = \frac{12 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega + (151)(15 \text{ k}\Omega)} = 4.96 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (150) (4.96 \text{ }\mu\text{A}) = 0.744 \text{ mA}$$

(b) V_E and V_B

$$I_E = \frac{V_E - (-12 \text{ V})}{15 \text{ k}\Omega}$$

 $\Rightarrow$

$$V_E = I_E (15 \text{ k}\Omega) - 12 \text{ V}$$

$$I_E = I_B + I_C = 4.96 \text{ }\mu\text{A} + 0.744 \text{ mA} = 0.748 \text{ mA}$$

Now

$$V_E = (0.748 \text{ mA}) (15 \text{ k}\Omega) - 12 \text{ V} = -0.78 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} - 0.78 \text{ V} = -0.08 \text{ V}$$

(c) V_C and V_{CE}

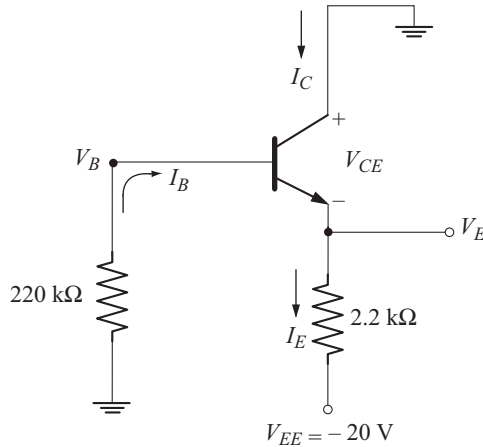
$$\begin{aligned}
 V_C &= V_{CC} - I_C R_C \\
 &= 18 \text{ V} - (0.744 \text{ mA})(10 \text{ k}\Omega) = 10.56 \text{ V} \\
 V_{CE} &= V_C - V_E = 10.56 \text{ V} - (-0.78 \text{ V}) = 11.34 \text{ V}
 \end{aligned}$$

Example 2.21

For the circuit shown below determine

- (a) I_B , I_C and I_E
 (b) V_B and V_E
 (c) V_C and V_{CE}

Assume silicon transistor with $\beta = 100$.

**Solution**

- (a) I_B , I_C and I_E

Applying KVL to base emitter circuit we have

$$-I_B (220 \text{ k}\Omega) - V_{BE} - I_E (2.2 \text{ k}\Omega) + 20 \text{ V} = 0 \quad (\text{A})$$

$$I_E = (1 + \beta) I_B = 101 I_B$$

Solving for I_B from Equation (A), we obtain

$$I_B = \frac{20 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega + (101)(2.2 \text{ k}\Omega)} = 43.64 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (100)(43.64 \text{ }\mu\text{A}) = 4.36 \text{ mA}$$

$$I_E = I_B + I_C = 43.64 \text{ }\mu\text{A} + 4.36 \text{ mA} = 4.4 \text{ mA}$$

(b) V_B and V_E

$$I_E = \frac{V_E - (-20 \text{ V})}{2.2 \text{ k}\Omega}$$

$\Rightarrow$

$$\begin{aligned} V_E &= I_E (2.2 \text{ k}\Omega) - 20 \text{ V} \\ &= (4.4 \text{ mA}) (2.2 \text{ k}\Omega) - 20 \text{ V} = -10.32 \text{ V} \\ V_B &= V_{BE} + V_E = 0.7 \text{ V} - 10.32 \text{ V} = -9.62 \text{ V} \end{aligned}$$

Alternatively

$$V_B = -I_B (220 \text{ k}\Omega) = -(43.64 \text{ }\mu\text{A}) (220 \text{ k}\Omega) = -9.6 \text{ V}$$

(c) V_C and V_{CE}

Since the collector terminal is grounded

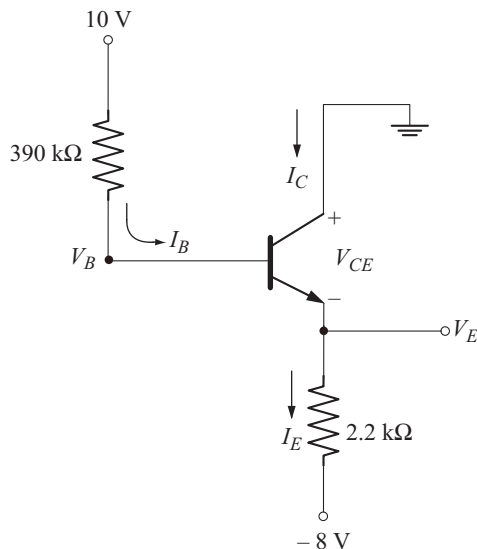
$$\begin{aligned} V_C &= 0 \text{ V} \\ V_{CE} &= V_C - V_E = 0 \text{ V} - (-10.32 \text{ V}) = 10.32 \text{ V} \end{aligned}$$

Example 2.22

For the circuit shown determine

- (a) I_B , I_C and I_E
- (b) V_B and V_E
- (c) V_C and V_{CE}

Assume silicon transistor with $\beta = 100$.



Solution(a) I_B , I_C and I_E

Applying KVL to base-emitter circuit, we have

$$10 \text{ V} - I_B (390 \text{ k}\Omega) - V_{BE} - I_E (2.2 \text{ k}\Omega) + 8 \text{ V} = 0$$

Using $I_E = (1 + \beta) I_B = 101 I_B$, we have

$$I_B [390 \text{ k}\Omega + (101) (2.2 \text{ k}\Omega)] = 17.3 \text{ V}$$

$$I_B = \frac{17.3 \text{ V}}{612.2 \text{ k}\Omega} = 28.25 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (100) (28.25 \text{ }\mu\text{A}) = 2.82 \text{ mA}$$

$$I_E = 101 I_B = (101) (28.25 \text{ }\mu\text{A}) = 2.85 \text{ mA}$$

(b) V_E and V_B

$$I_E = \frac{V_E - (-8 \text{ V})}{2.2 \text{ k}\Omega}$$

 $\Rightarrow$

$$V_E = I_E (2.2 \text{ k}\Omega) - 8 \text{ V}$$

$$= (2.85 \text{ mA}) (2.2 \text{ k}\Omega) - 8 \text{ V} = -1.73 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} - 1.73 \text{ V} = -1.03 \text{ V}$$

Alternatively

$$I_B = \frac{10 \text{ V} - V_B}{390 \text{ k}\Omega}$$

 $\Rightarrow$

$$V_B = 10 \text{ V} - I_B (390 \text{ k}\Omega)$$

$$= 10 \text{ V} - (28.25 \text{ }\mu\text{A}) (390 \text{ k}\Omega) = -1.02 \text{ V}$$

(c) V_C and V_{CE}

Since the collector terminal is grounded

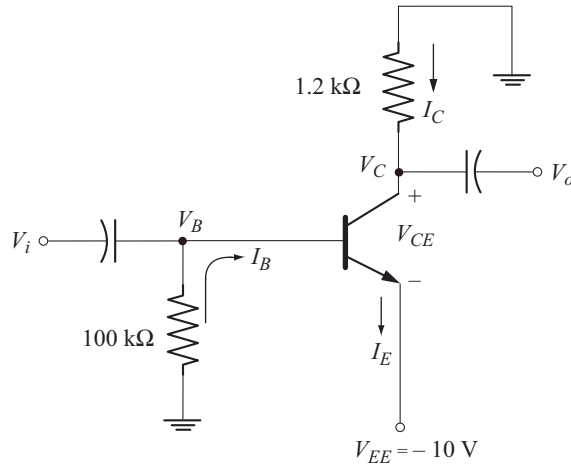
$$V_C = 0 \text{ V}$$

$$V_{CE} = V_C - V_E = 0 \text{ V} - (-1.73 \text{ V}) = 1.73 \text{ V}$$

Example 2.23

For the circuit shown below determine

(a) I_B , I_C and I_E (b) V_E and V_B (c) V_C and V_{CE} Assume silicon transistor with $\beta = 60$.



Solution

For dc analysis, all capacitors can be treated as open circuits.

(a) I_B , I_C and I_E

Applying KVL to base-emitter circuit, we have

$$-I_B (100 \text{ k}\Omega) - V_{BE} + 10 \text{ V} = 0$$

$$I_B = \frac{10 \text{ V} - 0.7 \text{ V}}{100 \text{ k}\Omega} = 93 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (60) (93 \text{ }\mu\text{A}) = 5.58 \text{ mA}$$

$$I_E = I_B + I_C = 93 \text{ }\mu\text{A} + 5.58 \text{ mA} = 5.67 \text{ mA}$$

(b) V_E and V_B

Since emitter terminal is connected to V_{EE} ,

$$V_E = V_{EE} = -10 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} - 10 \text{ V} = -9.3 \text{ V}$$

Alternatively,

$$V_B = -I_B (100 \text{ k}\Omega) = -(93 \text{ }\mu\text{A}) (100 \text{ k}\Omega) = -9.3 \text{ V}$$

(c) V_C and V_{CE}

$$I_C = \frac{0 \text{ V} - V_C}{1.2 \text{ k}\Omega}$$

$\Rightarrow$

$$V_C = I_C (1.2 \text{ k}\Omega) = (5.58 \text{ mA}) (1.2 \text{ k}\Omega) = 6.69 \text{ V}$$

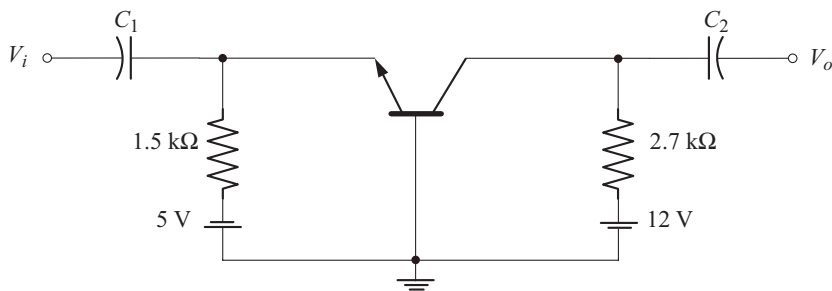
$$V_{CE} = V_C - V_E = 6.69 \text{ V} - (-10 \text{ V}) = 16.69 \text{ V}$$

Example 2.24

For the circuit shown below determine

- (a) I_B , I_C and I_E
- (b) V_B and V_E
- (c) V_C and V_{CE}
- (d) V_{CB}

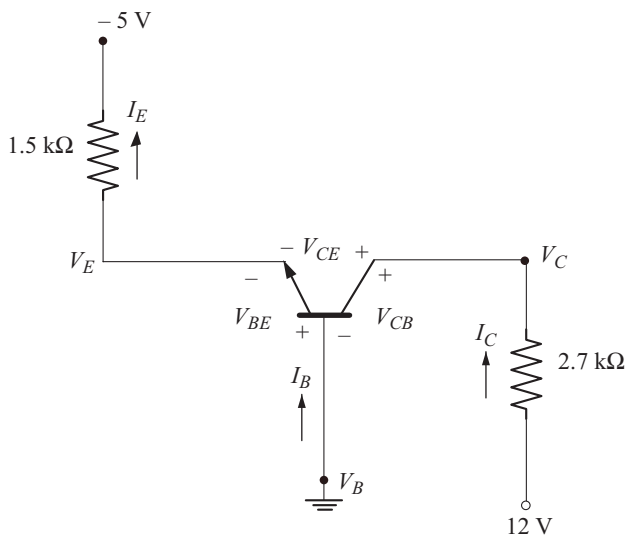
Assume silicon transistor with $\alpha = 0.98$.

**Solution**

In the given circuit, the transistor is in CB configuration.

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = 49.$$

Let us rewrite the given circuit in the convenient form as shown below.



(a) I_B , I_C and I_E

Applying KVL to base-emitter circuit, we have

$$-V_{BE} - I_E (1.5 \text{ k}\Omega) + 5 \text{ V} = 0$$

$$I_E = \frac{5 \text{ V} - 0.7 \text{ V}}{1.5 \text{ k}\Omega} = 2.86 \text{ mA}$$

$$I_B = \frac{I_E}{1 + \beta} = \frac{2.86 \text{ mA}}{50} = 57.2 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (49) (57.2 \text{ }\mu\text{A}) = 2.8 \text{ mA}$$

(b) V_B and V_E

Since the base terminal is grounded

$$V_B = 0 \text{ V}$$

$$I_E = \frac{V_E - (-5 \text{ V})}{1.5 \text{ k}\Omega}$$

$$\begin{aligned} V_E &= I_E (1.5 \text{ k}\Omega) - 5 \text{ V} \\ &= (2.86 \text{ mA}) (1.5 \text{ k}\Omega) - 5 \text{ V} = -0.71 \text{ V} \end{aligned}$$

It should be noted that, $V_E = -V_{BE}$

(c) V_C and V_{CE}

$$I_C = \frac{12 \text{ V} - V_C}{2.7 \text{ k}\Omega}$$

$\Rightarrow$

$$V_C = 12 \text{ V} - I_C (2.7 \text{ k}\Omega)$$

$$= 12 \text{ V} - (2.8 \text{ mA}) (2.7 \text{ k}\Omega) = 4.44 \text{ V}$$

$$V_{CE} = V_C - V_E = 4.44 \text{ V} - (-0.71 \text{ V}) = 5.15 \text{ V}$$

(d) V_{CB}

$$V_{CB} = V_C - V_B = 4.44 \text{ V} - 0 \text{ V} = 4.44 \text{ V}$$

Example 2.25

For the circuit shown below calculate

(a) I_B , I_C and I_E

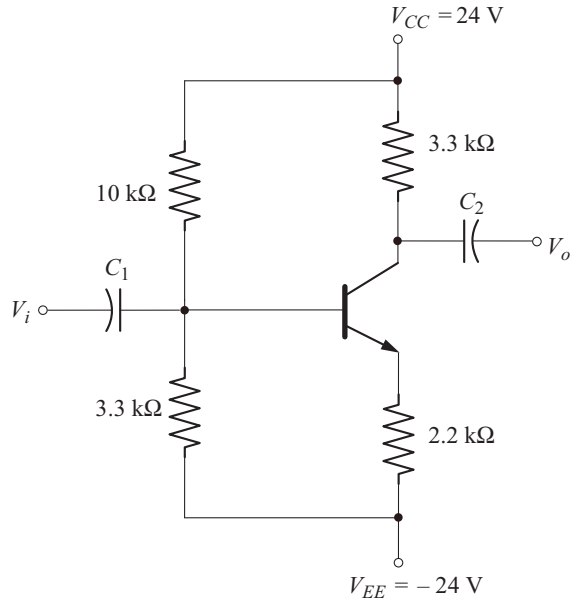
(b) V_E and V_B

(c) V_C and V_{CE}

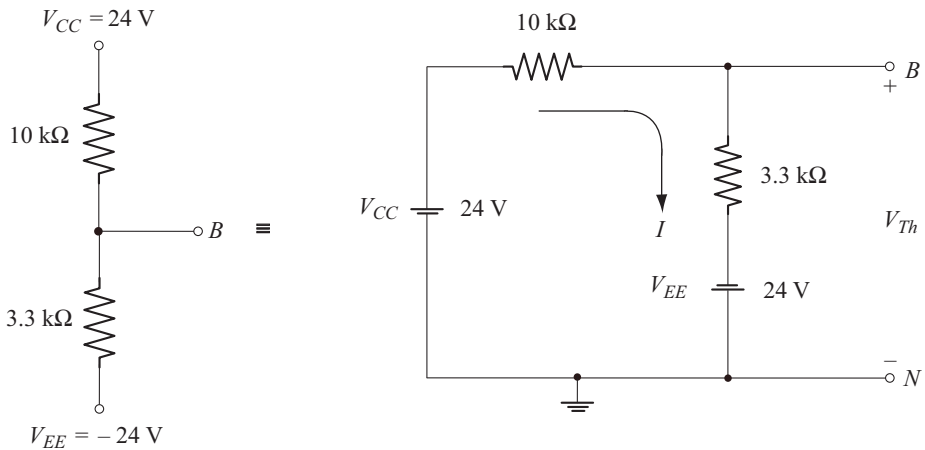
(d) V_{CB} and $I_{C(\text{sat})}$

(e) Draw the dc load line and indicate the Q point.

Assume silicon transistor with $\beta = 110$.

**Solution**

First let us obtain the Thevenin equivalent of the voltage divider network.

**Fig. A** Voltage divider network**Fig. B** Circuit to find V_{Th}

Applying KVL to the circuit of Fig. B we have

$$24 \text{ V} - I [10 \text{ k}\Omega + 3.3 \text{ k}\Omega] + 24 \text{ V} = 0$$

$$I = \frac{48 \text{ V}}{13.3 \text{ k}\Omega} = 3.61 \text{ mA}$$

also, $V_{Th} - I(3.3 \text{ k}\Omega) + 24 \text{ V} = 0$

$$\begin{aligned}
 V_{Th} &= I(3.3 \text{ k}\Omega) - 24 \text{ V} \\
 &= (3.61 \text{ mA})(3.3 \text{ k}\Omega) - 24 \text{ V} = -12 \text{ V}
 \end{aligned}$$

Negative sign implies that, base terminal is negative with respect to ground.

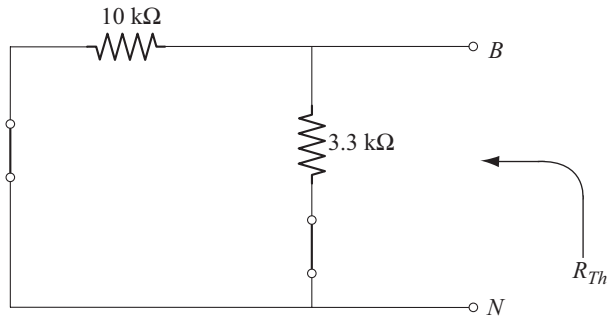


Fig. C Circuit to find R_{Th}

From the circuit of Fig. C.

$$R_{Th} = 10 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = 2.48 \text{ k}\Omega$$

The given circuit is redrawn in Fig. D with Thevenin equivalent inserted.

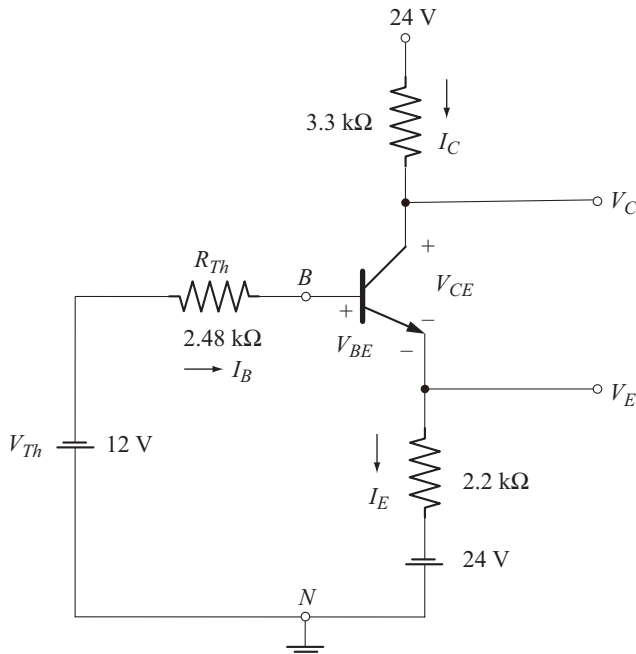


Fig. D

(a) I_B , I_C and I_E

Applying KVL to the base emitter circuit of Fig. D we have

$$-12 \text{ V} - I_B (2.48 \text{ k}\Omega) - V_{BE} - I_E (2.2 \text{ k}\Omega) + 24 \text{ V} = 0 \quad (\text{A})$$

But $I_E = (1 + \beta) I_B = 111 I_B$.

Using this in Equation (A), we have

$$\begin{aligned} I_B [2.48 \text{ k}\Omega + (111) (2.2 \text{ k}\Omega)] &= 12 \text{ V} - V_{BE} \\ I_B &= \frac{12 \text{ V} - 0.7 \text{ V}}{2.48 \text{ k}\Omega + (111)(2.2 \text{ k}\Omega)} = 45.8 \text{ }\mu\text{A} \\ I_C &= \beta I_B = (110) (45.8 \text{ }\mu\text{A}) = 5.03 \text{ mA} \\ I_E &= I_B + I_C = 45.8 \text{ }\mu\text{A} + 5.03 \text{ mA} = 5.07 \text{ mA} \end{aligned}$$

(b) V_E and V_B

$$\begin{aligned} I_E &= \frac{V_E - (-24 \text{ V})}{2.2 \text{ k}\Omega} \\ \Rightarrow V_E &= I_E (2.2 \text{ k}\Omega) - 24 \text{ V} \\ &= (5.07 \text{ mA}) (2.2 \text{ k}\Omega) - 24 \text{ V} = -12.84 \text{ V} \\ V_B &= V_{BE} + V_E = 0.7 \text{ V} - 12.84 \text{ V} = -12.14 \text{ V} \end{aligned}$$

Alternatively, from the circuit of Fig. B

$$V_B = V_{BN} = V_{Th} = -12 \text{ V}$$

(c) V_C and V_{CE}

$$\begin{aligned} I_C &= \frac{24 \text{ V} - V_C}{3.3 \text{ k}\Omega} \\ \Rightarrow V_C &= 24 \text{ V} - I_C (3.3 \text{ k}\Omega) \\ &= 24 \text{ V} - (5.03 \text{ mA}) (3.3 \text{ k}\Omega) = 7.4 \text{ V} \\ V_{CE} &= V_C - V_E = 7.4 \text{ V} - [-12.84 \text{ V}] = 20.24 \text{ V} \end{aligned}$$

(d) V_{CB} and $I_{C(\text{sat})}$

$$V_{CB} = V_C - V_B = 7.4 \text{ V} - (-12.14 \text{ V}) = 19.54 \text{ V}$$

Applying KVL to collector circuit of Fig. D we have

$$24 \text{ V} - I_C (3.3 \text{ k}\Omega) - V_{CE} - I_E (2.2 \text{ k}\Omega) + 24 \text{ V} = 0 \quad (\text{B})$$

When transistor is in saturation

$$V_{CE(\text{sat})} \approx 0 \text{ V}$$